

Fort Larned National Historic Site

1767 Kansas Highway 156 Larned. Kansas

Roof Surface Replacement of Visitor Center/Administration

Building

PMIS 312839

Drawing No:

PROJECT SPECIFICATIONS



NATIONAL PARK SERVICE

2024-12-19

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DIVISION 1 SHORT FORM GENERAL REQUIREMENTS

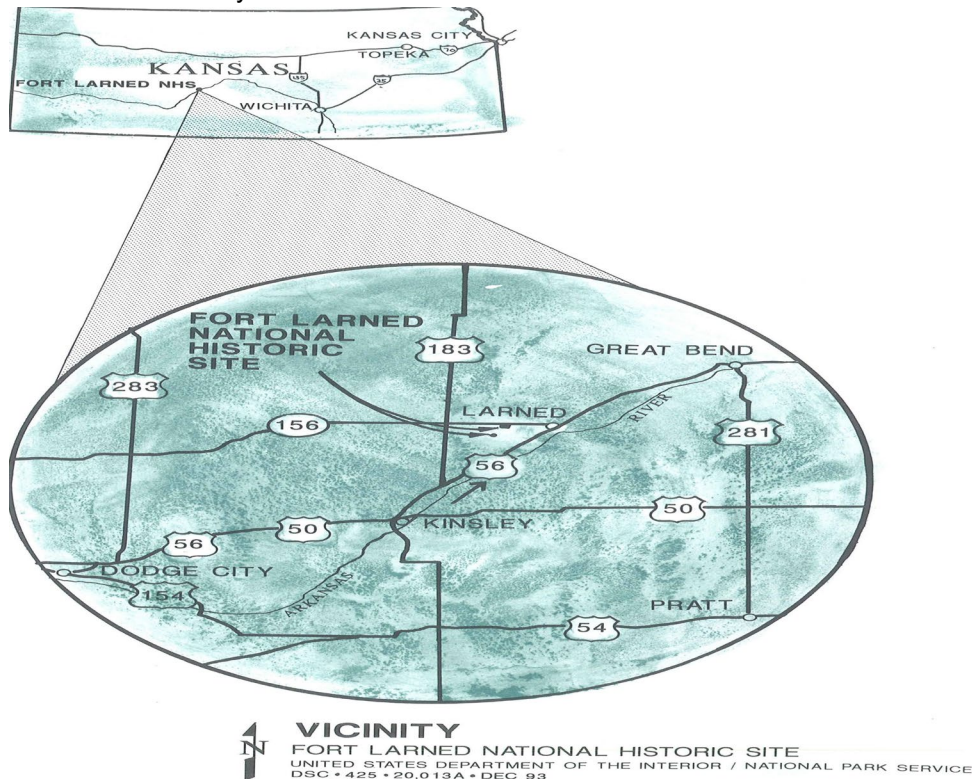
PART 1 – GENERAL

ITEM 1 - DESCRIPTION OF WORK

1.1 Background: Fort Larned served as US military outpost to protect US Mail service from 1859 to 1880's before being decommissioned and sold. It served as a ranch and farm until it became a National Historic Site in 1968 and is managed by the National Park Service. The fort consists of 9 original structures from the 1800's and one reconstructed building. This contract work is focused upon the Visitor Center/ Administration building. This building is in the North-northwest corner of the parade ground. The structure is a single-story T-shape stone structure that measures approximately 150-foot, x 40-foot with a 35-foot by 30-foot centered addition and a covered porch approximately 11-foot wide and the length of the building. The roof penetrations include five brick chimneys, lightning protection system and antennas for security cameras. Lightning protection system is attached to roof and chimneys, with grounding connections.

1.2 Work Covered by the Contract Documents:

A. Project Location: Fort Larned National Historic Site 1767 Kansas State Highway 156, Larned Kansas. Worksite is accessible from 180th avenue service entrance into the park. Image 2 Provides the Vicinity of Fort Larned NHS.



(Image 2) Vicinity Map

- B. Work consists of:
 - a. Remove approximately 10538 Square Feet of western red cedar shingles.
 - b. Inspect and replace an estimated 15% of the 10538 SF of skip sheeting. Matching dimensional size and wood species.
 - c. Install approximately 10538 SF of fire retardant western red cedar wood shingles, hand nailing the shingle installation.
 - d. Detach the lightning protection system that will be affected with removal of the roof surfaces. Reinstallation will be by a certified UL listed installer and recertification is required.
 - e. Install debris containment (traps) in the attic space of structure and remove the traps and all debris from the attic space.
 - f. Contractor must provide safe access for the government and public to the structure during the entire project period.
 - g. Install new valley flashing and step flashing at each shingle course at all roof penetrations. Apron and counter flashing may be reused if undamaged and will be replaced if damaged. Use existing reglets in masonry joinery (mortar joints). Do not cut reliefs into masonry units.
 - h. See Attached Hover Document for all approximate quantities and dimensions.
- C. All work will be performed under a single contract.
- D. Period of performance: Work shall be completed within 120 calendar days of the start date provided on the Notice to Proceed.
 - a. Clarification: The Notice to Proceed will be issued with a period of performance that is all inclusive of supply lead times, material deliveries, decommissioning, installation, anticipated average weather days for location, final commissioning, and acceptance of completed work. The contractor will be expected to provide a project schedule with adequate critical path details to ensure all work is completed within the stated calendar days from issuance of the Notice to Proceed. If weather impacts project beyond typical average weather patterns, contractor shall provide documentation in request for additional days.

1.3 Existing Condition Protections:

- A. Preservation of Historical and Archeological Fabric:
 - i. The Historic and Archeological Data Preservation Act of 1974 provides for the preservation of historical and archeological data that might otherwise be lost as the result of alterations to the terrain caused by a federal or federally licensed activity or program.
 - ii. IF, in connection with operations under this contract, the Contractor, subcontractors, or employees of any of them, discovers, encounters, or becomes aware of any possible historical or archeological remains or data, objects or sites of cultural value on the project area, such as historical ruins, graves or grave markers, fossils, or artifacts not specifically noted within the contract. The Contractor shall immediately suspend all operations in the vicinity and continue work in other areas outside a 50-foot radius of discovery and notify the Contracting Officer immediately. When safe to do so, follow up in writing with photographs within 2 days, giving the location and nature of the findings. No

- objects of cultural resource value may be removed. Contracting Officer will contact other entities as required.
- iii. Where appropriate by reason of discovery, the Contracting Officer may order delays in the time of performance and/or changes in the work. If such delays and/or changes are ordered, the time of performance and contract price shall be adjusted in accordance with Changes clause.
 - iv. The Contractor will be responsible for protecting the cultural resources within and adjacent to the project area from damage. In addition, the Contractor will be liable for all damage to the cultural resources caused by their actions or the actions of their agents or representatives. The Contractor shall immediately notify the Contracting Officer or their representative if any damage occurs to any cultural resource and immediately suspend work in the area in which damage has occurred until authorized to proceed.
 - v. The Contractor agrees to include this language within all subcontracts.
- B. The structure involved in the work of this Contract is a designated historic structure listed in the National Register of Historic Places. Every precaution shall be taken by the Contractor to prevent damage to existing historical materials, structural components of the building and to the surrounding area. Should any damage be incurred as a result of the Contractor's operations at the site, it shall be repaired, or the damaged materials replaced in kind by the Contractor to the satisfaction of the Contracting Officer at no additional expense to the Government.
- C. Protection of Landscaping and Surrounding Structures: Coordinate schedules as required. The Contractor shall take all necessary measures to provide protection for all trees, fencing, walks and other landscape features and surrounding structures near the work site as noted with the contract. Any such surrounding features damaged, scarred or broken as a result of the Contractor's operations at the site outside of agreed upon documented extents, shall be treated, repaired or replaced in kind to the satisfaction of the Contracting Officer at no additional expense to the Government.

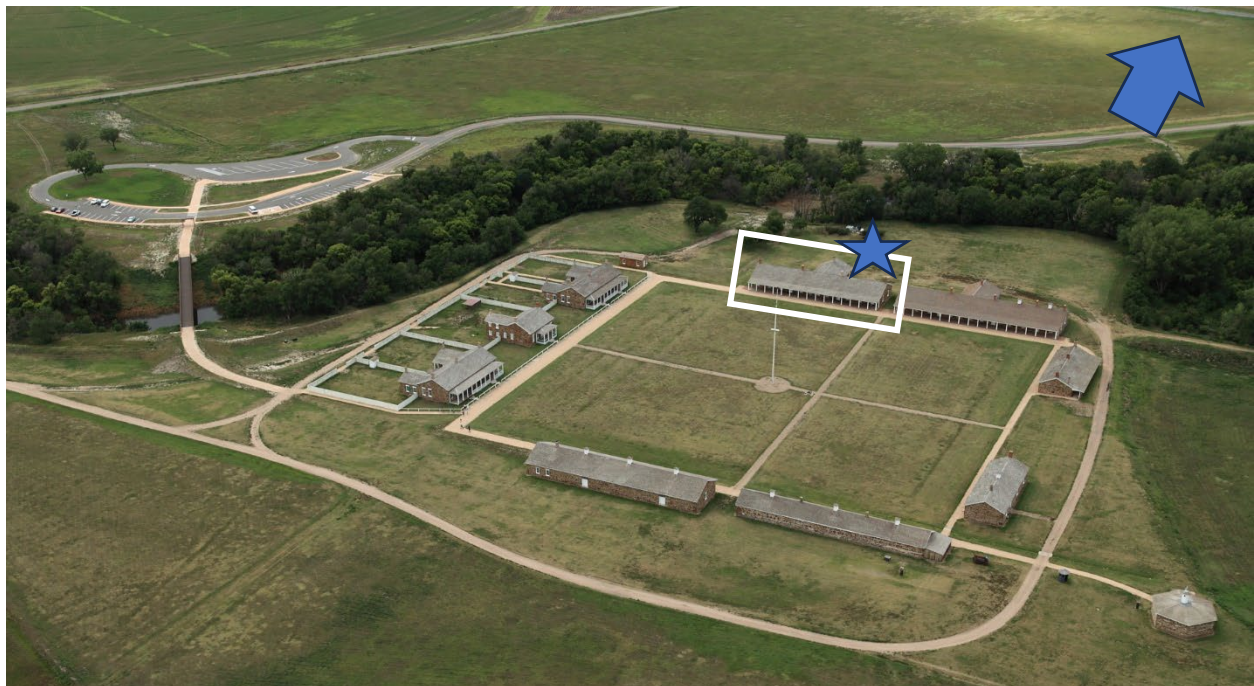
ITEM 2 – PROTECTING EXISTING UTILITIES

- 2.1 Protecting Existing Utilities: Notify Contracting Officer (CO), Contracting Officer's Representative (COR) and utility companies as required to request location services prior to and to coordinate any excavation/digging operations, and/or for demolition if within Scope of Work.
- 2.2 Contractor shall be responsible for locating and preventing damage to known utilities on site or indoors. If contractor damage occurs, alert CO and COR as soon as safely possible, and repair utility at no additional expense to the Government. If damage occurs to an unknown utility, contact CO and COR as soon as safely possible.
- 2.3 Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by Government or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated below:
- A. Notify CO and COR not less than two business days in advance of proposed utility interruptions.

B. Hours for Utility Shutdowns: Shutdowns shall not exceed 2 to 3 hours. Temporary provisions will be needed by the Contractor for periods longer than stated allowance or if required for critical systems as determined by COR.

ITEM 3 - CONTRACTOR'S USE OF PREMISES

- 3.1 Conduct of Operations: At all times the contractor shall conduct his operations in conformance with the rules and regulations promulgated by the Secretary of the Interior for the National Park Service, and applicable park rules and regulations prescribed by the Park Superintendent.
<https://www.nps.gov/fols/learn/management/superintendent-s-compendium.html>



(Image 3) White outline is confine of work area. Star Shape, indicate staging area for contractor use. Staging area is limited to a 20-foot by 50-foot area within the confines of work area.

- A. Communications: Contractor shall provide the Contracting Officer and COR with methods of communications including land and/or cell phone numbers (note some areas of work have limited-service coverage), email and in case of emergency 24-hour contact options for duration of time operations.
- B. No signs or advertisements (except those specified herein and as required for safety) shall be displayed on the construction site or within the park unless approved by the Contracting Officer.
- C. Smoking is not permitted within any Federal building or within 25 feet of entrances, operable windows, or outdoor air intakes.

- D. Public Use of Site: Contractor shall at all times conduct operations to ensure the least inconvenience to the public. Contractor must provide safe access for the government and public to the structure during the entire project period. Limits: Contractor is limited to 1 vehicle within the park's historic corridor, all other vehicles are to be park in the maintenance or in visitor parking lots. Staging area is limited to Northeast elevation of the building. Contractor must be off government property by 4PM.

- 3.2 On-site work hours: Work shall be generally performed during normal business working hours of 7AM to 4PM. Monday through Friday, except when otherwise indicated. Work on Saturdays, Sundays, Federal holidays, early morning, or at night may not be performed without prior consent from the CO and COR. Submit requests for work outside of the stated normal business hours to the CO and COR for approval at least 48 hours in advance of the work.

The following Federal holidays are observed by the National Park Service:

New Year's Day
Inauguration Day
Martin Luther King, Jr's Birthday
Washington's Birthday
Memorial Day
Juneteenth National Independence Day
Independence Day
Labor Day
Columbus Day
Veterans Day
Thanksgiving Day
Christmas Day

- Note: Any of the above holidays falling on a Saturday will be observed on the preceding Friday; holidays falling on Sunday will be observed on the following Monday. The specific day that each holiday is recognized for each year can be found at the following site: [Federal Holidays \(opm.gov\)](https://www.opm.gov/federal-holidays/).

3.3 Contractor Use of Site:

- A. General: Contractor shall have limited use of the site for construction operations. Limit use of premises to construction area as shown on site plan. Do not disturb portions of Project site beyond areas in which the Work is indicated. Contractor is limited to 1 vehicle within the park's historic corridor, all other vehicles are to be park in the maintenance or in visitor parking lots. Staging area is limited to Northeast elevation of the building. Contractor must be off government property by 4PM. See site plan in Appendix A.
- B. Driveways and Entrances: Keep driveways, access roads, and entrances serving premises clear and available to Government, Government's employees, and emergency vehicles at all times. Do not use these areas for parking or storage of materials. Schedule deliveries to minimize use of driveways and entrances.
- C. Do Not disturb site outside of noted disturbance area, including earthwork, and clearing of vegetation, to:

1. a. 20 feet beyond building perimeter except on the South elevation which is limited to 6 feet.
b. Contractor must provide safe access for the government and public to the structure during the entire project period.
c. Do not place or stage material, equipment, or vehicle near emergency exit route, on top of Septic system or water distribution access covers.
d. Keep waste containers within staging area.

Contractor shall provide appropriate mats for equipment on turf areas to reduce soil compaction. Contractor shall remediate soil compaction as directed to do so by the CO/COR.

D. Utilities

1. General: Cost or use charges for temporary facilities shall be included in the Contract Sum as required.
2. Water Service: Water from existing water system is available for use without metering and without payment of use charges. Provide connections and extensions of services as required for construction operations without any additional permit costs. Use resources wisely and minimize waste of water.
3. Electric Power Service: Electric power from existing system is available for use without metering and without payment of use charges. Provide connections and extensions of services as required for construction operations without any additional permit costs. Power may or may not meet your construction power tool requirements, provide generators if questionable. Use power in a resourceful manner and avoid excess power waste.
4. Sanitary Operations:
 - a. Toilets: Contractor shall be able to use the existing park restroom facilities, specific location to be discussed during the preconstruction meeting. The restroom facilities shall be kept in immaculate condition. At the discretion of Park personnel and/or COR; contractor may lose the use of the restroom facilities if they are not kept clean and orderly. Restroom use will be a "One strike and you're out" policy. If restroom privileges are revoked Contractor will be required to provide a toilet for contracting personnel, sufficiently lighted and ventilated toilet facilities in weatherproof, sight proof, handicap accessible (if required), sturdy enclosures with privacy locks. Coordinate with COR for approved location.

E. Security and Protection facilities:

1. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with state and federal environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.
2. Cleaning of Equipment: The Contractor shall ensure that prior to moving on to the Project Area, all equipment, is free of soil, seeds, vegetative matter, or other debris that could contain or hold seeds. Ensure that all equipment has been pressure washed and is free of exotic species prior to start-up of operations and moving of equipment to Project Area. Equipment shall be considered free of soil, seeds, and other debris when a visual

inspection does not disclose such material. Disassembly of equipment components or specialized inspection tools are not required.

3. Tree and Plant protection is not required for this project.
4. Pest Control: Follow best practices to minimize attraction and harboring of rodents, roaches, and other pests and to perform extermination and control procedures at regular intervals so Project will be free of pests and their residues at project completion. Perform control operations lawfully, using environmentally safe materials.
5. Temporary Enclosures: Provide and maintain temporary enclosures such as snow fence as required for protection of construction in progress and from access by visitors or staff, other construction operations, and similar activities.
6. Hot Work Permit: Submit HW-1 Hot work permit form and receive approval from COR prior to Hot work activities commencing on site. <https://www.nfpa.org/downloadable-resources/forms/hot-work-permit>

3.4 Storage - Staging Areas:

- A. Confine Storage of materials and equipment to the project staging area noted on the plans/aerial. Specific storage units shall be provided by the contractor and exact location within staging area shown in appendix A below may be discussed at the preconstruction briefing. All storage units provided by the contractor shall be in a clean and acceptable condition to the COR. Security of the materials and equipment on site is the Contractor's responsibility.

3.5 Preservation of Natural Features:

- 3.6 Hauling Restrictions: Contractor shall verify and be responsible for the path accessing the site and obtaining / maintaining all local, state and federal permits/licenses associated with hauling/road requirements and dumping of materials associated with the project.

3.7 Waste Removal, Recycling, and General Housekeeping:

- A. Keep project site in a neat, orderly, and safe condition at all times.
 1. Contractor is responsible for providing dumpsters for waste and recycling and for removing all waste created by the project. Contractor shall recycle to the highest degree possible and empty dumpsters as necessary to avoid debris on site. Coordinate with COR for approved dumpster locations prior to commencing construction.
 2. Dispose of waste materials and recycling legally outside the park. The contractor shall not use park dumpsters.
 3. Address Recycling: Government goal is to salvage and recycle as much nonhazardous construction waste as possible. Use sustainable methods to the extent possible during this project, recycling material as much as possible within local

means. Maintain accurate documentary records for submittal of all recycled materials used including fly ash and slag cement.

4. For all waste and recycling removals offsite, contractors shall provide estimates and/or tickets to quantify material weights and report back to the park, submit to COR for review and record filing.

3.8 Occupancy Requirements for Buildings:

A. Existing Buildings:

1. Full Government Occupancy: Government will occupy buildings that will be under construction during the entire contract period. Cooperate with Government during construction operations to minimize conflicts and facilitate Government usage. Perform the Work so as not to interfere with Government's day-to-day operations. Maintain existing exits, unless otherwise indicated. Contractor will maintain access to existing walkways, corridors, and other adjacent occupied or used facilities. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from the CO or COR. Contractor will maintain existing building in a weather tight condition throughout construction period. Repair damage caused by construction operations. Protect building and its occupants during construction period.

B. New /Existing Buildings Unoccupied: N/A

3.9 Photo Documentation:

- #### A. Contractor must provide existing condition, progress, and completion photos. All photos shall be in focus, show maximum range of depth of field general photo and then close up detail photo as needed. Shall be in JPEG digital form and submitted preferably electronically by email or optional via CD-R Archival Gold or DVD-R Archival Gold media with printed labels and 8 mega pixels minimum within 7 days from time of taken or as approved by COR.

B. Existing Condition Images:

1. Before starting construction, take color, dated, digital images of project site and adjacent property/roads/etc. that may be impacted by construction activity/traffic. Include existing items to remain or be salvaged from different vantage points and with context for location identification.

C. Construction Progress Images:

1. Document all sub-surface work/ trenches etc. prior to backfill with photos covering enough overall field view to show location and then zoom in for details. (Locations and dimensions shall also be noted on as built drawing set.)
2. Maintain organized index for photos with date, description, etc.
3. Additional images as needed when questions or emergency situations occur. Submit within 24 hours.

D. Completion Images:

1. Document all completed work within scope of work.
2. Maintain organized index for photos with date, description, etc..
3. Additional images as needed when questions or emergency situations occur. Submit within 24 hours.

3.10 Fire and Life Safety:

- A. As a minimum, conduct one weekly 15 minute "toolbox" safety meetings. These meetings shall be conducted by a foreman or supervisor and attended by all construction personnel at the worksite. Topics need to coincide with work scheduled for the following week. Document and submit meeting minutes to the CO and COR within one day after the meeting and document the meetings on the daily reports.
- B. Comply with the requirements of NFPA 241 (Standard for Safeguarding Construction, Alteration, and Demolition Operations). Take all necessary precautions to prevent fire during construction. Provide adequate ventilation during use of volatile or noxious substances.
- C. Store and handle hazardous materials in accordance with manufacturers and OSHA 29CFR1926 Subpart D requirements [1926 | Occupational Safety and Health Administration \(osha.gov\)](https://www.osha.gov). Maintain readily available, on site, MSDS/Safety Data Sheets (SDS) for each chemical.
 1. Immediately report all spills of hazardous materials to the CO, COR, and park.
 2. Maintain a spill emergency response kit.
 3. Train employees how to respond to a spill and use the emergency response kit.

3.11 Noise and Acoustics Management

- A. Contractor shall not have personal radios or other media sources and they shall not be broadcast within NPS property.
- B. Noise Control: Perform operations to minimize noise. Perform noise-producing work in less sensitive hours of the day or week as directed by the Contracting Officer or COR.
- C. Repetitive and/or intermittent, high-level noise: Permitted only during Daytime.

ITEM 4 - COMMENCEMENT, PROSECUTION, AND COMPLETION

- 4.1 The Contractor shall commence work in accordance with the terms and conditions of the contract. Contractor shall coordinate and incorporate long lead and delivery time items within the schedule.

ITEM 5 - EXAMINATION OF THE SITE

- 5.1 All bidders are urged to visit the job site and examine existing conditions before submitting bids. Contractors will be accountable for understanding existing conditions. No cost changes for items visible during site visit will be allowed.

ITEM 6 - ABBREVIATIONS

- 6.1 The list of abbreviations, standards and organizations which may be used in/referred within the Construction Specifications or Special Provisions, may be found linked from the listed page below. On the home page, follow the link under “**Contracting and Procurement**” labeled “**NPS Reference Standards.**” <https://www.nps.gov/orgs/1671/do-business-with-us.htm>

ITEM 7 - ACCIDENT PREVENTION

- 7.1 Description: The work of this section consists of establishing and implementing an effective accident prevention program (APP) and providing a safe environment for all personnel and visitors.
- 7.2 Submittals:
- A. Accident Prevention (Safety) Plan (APP): Submit a **site-specific** accident prevention (safety) plan. The Plan shall be written to comply with OSHA and project requirements (a generic plan is not acceptable) including but not limited to the following:
1. Name and qualifications of responsible supervisor to carry out the program.
 2. First aid and rescue procedures.
 3. Training, both initial and continuing.
 4. Outline of each phase of the work, the hazards associated with each major phase, and the methods proposed to provide for property protection and safety of the public, National Park Service personnel, and Contractor's employees. Identify the work included under each phase, with a Job Hazard Analysis (JHA)/Job Safety Analysis (JSA), etc. A sample JHA template may be provided upon request.
 5. Planning for possible emergency situations, such as cave-ins, earthquake, explosions, fires, floods, power outages, slides, and windstorms. Such planning shall take into consideration the nature of construction, site conditions, and degree of exposure of persons and property.
 6. Contractor to identify nearest location of medical facility for emergencies
 7. Recognition that works will be performed in remote areas with limited communication or cell phone coverage if applicable.
 8. Supplemental hazard specific plans may be required IAW 29 CFR 1926 or 1910.
- B. Meeting and Inspection Reports: Safety Meetings: As a minimum, conduct 15 minute "toolbox" safety meetings. These meetings shall be conducted by a foreman and attended by all construction personnel at the worksite. Submit a report of safety meetings and inspections to CO and COR within one day after the meeting and document the meetings on the daily reports.
- 7.3 Qualification of Employees:
- A. Ensure that employees are physically qualified to perform their assigned duties in a safe manner to include operators of vehicles and equipment. Provide operating instructions for all equipment.
- B. Do not allow employees to work whose ability or alertness is impaired because of drugs, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury.

- 7.4 Accident Reporting: The Contractor shall report all accidents to the CO and COR as soon as safely possible and assist the CO and COR and other officials as required in the investigation of the accident.
- 7.5 First Aid Facilities: Provide access to facilities for the number of employees and appropriate to the hazards associated with the types of ongoing construction work at the site.
- 7.6 Emergency Instructions: Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at the work site.
- 7.7 Personal Protective Equipment (PPE): It is the Contractor's responsibility to require all those working on or visiting the site to wear necessary protective equipment at all times.
 - A. PPE must meet requirements of applicable ANSI standards. Selection shall conform to OSHA 29CFR 1926 Subpart E.
 - B. A hard hat use area shall be designated by the Contractor. The hard hat area shall be posted by the Contractor in a manner satisfactory to the Contracting Officer.
 - C. All PPE will be serviceable at all times. At a minimum, maintain 2 sets of PPEs including hard hats and all other Accident Prevention Plan required equipment.
- 7.8 Hazardous Materials: It shall be the responsibility of all operators and contractors, functioning within the park to properly cleanup, mitigate and remediate, if necessary, all unauthorized discharges of hazardous materials or non-hazardous chemical and biological products released from fleet and/or other support vehicles or stationary sources. Response shall be consistent with guidelines established by federal, state, and local regulations, and as outlined within the operator's Safety or Hazardous Materials Business Plan.
 - A. If a spill, leak, or other release occurs, the operator shall as soon as possible, without impeding cleanup, notify the local 911 Emergency Response, COR and or the National Park Service Emergency Dispatch.

ITEM 8 - PRECONSTRUCTION CONFERENCE

- 8.1 If the Contracting Officer decides to conduct a preconstruction conference, the successful offeror will be notified and will be required to attend. The Contracting Officer's notification will include specific details regarding the date, time, and location of the conference, any need for attendance by subcontractors, and information regarding the items to be discussed. See Item 9.1C below.

ITEM 9 - SUBMITTALS

- 9.1 General Contracting Submittal Procedures:
 - A. General: Prepare and submit submittals required by individual Technical Specification Sections. Types of submittals are indicated in individual specific sections.

1. CM-16 Transmittal Form: All submittals shall be transmitted via electronic email using National Park Service form CM-16 form. The form will be provided by the Contracting Officer. No action will be taken on a submittal item unless accompanied by the CM-16 transmittal form. Submit completed CM-16 forms to the CO and the COR
- B. Submittal List: administrative required submittals are listed below in 9.1.C and does not include technical section submittal requirements included in each technical section.
- C. Preconstruction Conference Submittal: Satisfactory evidence of liability insurance coverage and workman's compensation, and any applicable bonds, must be submitted within 10 calendar days of Notice of Award. The following items are to be submitted prior to the Preconstruction Conference or as specified at the CO's discretion. The timeline for the documents to be submitted will be addressed at the pre-construction conference for those items that were not received and approved prior. Notice to Proceed may not be issued, and the Contracting Officer will consider other contractual remedies if not resolved as directed. Work shall not commence until written Notice to Proceed has been issued. Contracting Officer will notify Contractor of tentative date for the Pre-Construction Conference:
1. Letter designating Project Superintendent.
 2. Construction Schedule. (Must include shop drawings, procurement time including long lead time items, mobilization, trade work progression and completion at minimum.)
 3. A comprehensive breakdown of the Schedule of Values.
 4. Accident Prevention (Safety) Plan (see section 7 of this document).
 5. A list of Subcontractors for this project.
 6. Written statements from subcontractors certifying compliance with applicable labor standard clauses (SF1413).
 7. Quality Control Plan.
- 9.2 Project Submittal Process (Shop Drawings, Product Data, and Samples): Contracting Officer will provide sample cover sheet and a list of required submittals is included in each technical specification or attached at the end of this document.
- A. Shop Drawings: Indicate all coordinating materials and work impacting the installation coordination, connection details, anchorage requirements, hardware locations, and installation details and locations.
1. Submit Shop Drawings according to Part 9.1 of this section to include one digital file (.PDF or .TIF) and two full size hardcopy documents to COR.
- B. Product Data: Provide component construction and specification data sheets, anchorage requirements and any other product specific data.
1. Submit Product Data according to Part 9.1 of this section to include one digital file (.PDF or .TIF) to COR.
- C. Samples: Submit 2 color samples 4" x 6" minimum in size for all product color selections. Paint samples shall be actual paint finish drawdown for all project colors.

1. Submit Samples CM-16 Transmittal Form according to Part 9.1 of this section to include one digital file (.PDF or .TIF) and physical samples of actual material to COR.
 - D. For submittals specified above, forward submittals to Contracting Officer and COR at least 15 days before need for approval (depends on length of project).
 - E. After approving submittals, Contracting Officer and/or COR will return one copy to the Contractor. If submittals are not approved, Contracting Officer will return all copies to Contractor with reasons for rejection. Resubmit, identifying changes. Any work done before approval shall be at Contractor's own risk.
- 9.3 Approved Equals: All brand names noted are subject to 'or equal' products. If an item in this solicitation is identified as "brand name or equal" or lists more than 1 manufacturer, the description of the proposed item must reflect the characteristics and level of quality that will satisfy the Government's needs as demonstrated by the salient physical, functional, or performance characteristics that "equal" products must meet and/or exceed that as specified in the solicitation or relevant specification. Contractor shall incorporate and be responsible for all collateral costs due to the or equal proposed as required to provide a complete project.
- A. For each item proposed as an "approved equal," submit supporting data, including:
 1. Drawings and samples as appropriate.
 2. Comparison of the characteristics of the proposed item with that specified and with the salient characteristics provided.
 3. Changes required in other elements of the work because of the substitution.
 4. Name, address, and telephone number of vendor.
 5. Manufacturer's literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service, and state source of replacement materials.
 - B. A request for approval constitutes a representation that General Contractor:
 1. Has investigated and shall provide documentation in request showing that the proposed item is equal or superior in all respects to the salient characteristics specified.
 2. Will meet or exceed warranties for the proposed item specified.
 3. Has determined that the proposed item is compatible with interfacing items.
 4. Will coordinate the installation of an approved item and make all changes required in other elements of the work because of the substitution.
 5. Waives all claims for additional expenses that may be incurred as a result of the substitution.
 - C. Approved Equals will be evaluated and approved/disapproved during the submittal process.
- 9.4 Schedule of Values: Submit the Schedule of Values a minimum of one week prior to the Preconstruction Conference for COR review. Schedule of Values shall include defined work line items with cost break downs including labor, material, and equipment for COR approval.
- 9.5 Progress Reporting: The initial Construction Schedule and Schedule of Values are required a minimum of one week prior to the Preconstruction Conference. After the Notice to Proceed is

issued and after on-site work begins, progress will be documented via daily logs and by providing an updated Construction Schedule and Schedule of Values when requested, but minimally with each pay application. Submit digital copy electronically (PDF preferred) of all required documents. Construction Schedules are normally in bar chart form and indicate estimated starting and completion dates for each part of the work. For a pay application to be reviewed and processed in a timely manner, the following information must either be submitted prior to or accompany submission of the pay application:

- A. Daily/weekly progress reports during construction as part of the administration process.
- B. Updated Schedule of Values: Breakdown each lump-sum item into component work activities used in the schedule, for which progress payments may be requested. The Schedule of Values will form the basis for payment. The work activities broken out within the schedule of values shall be integrated into and made a logical part of the construction baseline schedule submitted under this specification. The total costs for the component work activities shall equal the contract price for that lump-sum item. The Contracting Officer may request data to verify accuracy of dollar values. A sample Schedule of Values format may be provided upon request.

ITEM 10 – TEMPORARY SERVICES

- 10.1 It is the Contractor's responsibility to provide temporary services, equipment, tools, and skilled labor as required for the chosen means and methods to perform the work in accomplishing this Contract.

ITEM 11 - PROJECT CLOSEOUT

- 11.1 Description: The work of this section consists of final cleanup, closeout submittals, final inspection procedures working toward Final Acceptance.
- 11.2 Cleaning: Remove all tools, equipment, surplus materials, and rubbish. Move attic stock to location coordinated with COR. At time of final inspection, project shall be thoroughly clean and ready for intended use. Inspector holds the right to reschedule if project is not ready for inspection.
- 11.3 Completion and Final Inspection: Contractor shall submit written certification of project completion and request a final inspection to the Contracting Officer and COR. The Contracting Officer or designee will make an inspection within 10 days of receipt of request or at a mutually agreeable time.
 - A. If the work is determined to be complete following the final inspection with minor comments, the COR will prepare and provide the Contractor a Punch List noting outstanding items to be addressed.
 - B. If the work is not determined to be complete following the final inspection, the Contracting Officer will notify Contractor in writing, stating reasons. Contractor, after completing the work according to the Contract Documents, shall resubmit completion and request a new final inspection. Re-inspection costs may be charged against the Contractor in accordance with the Inspection of Construction contract clause.

- C. Contractor shall complete the Punch List within 30 calendar days, or a time directed by the CO.
 - D. Once Contractor completes all items of work on the Punch List and all contractually required items are accepted, Contracting Officer will issue Letter of Final Acceptance of Work and request the final pay application.
 - E. If the Contractor fails to complete the work within the time frame, the Contracting Officer may correct the work with an appropriate reduction in contract price or charge for re-inspection costs in accordance with the Inspection of Construction contract clause.
 - F. Disposition of any permits and warranties required by the specifications in this section.
- 11.4 Final Acceptance of the Work: After all punch list deficiencies have been corrected, a Letter of Final Acceptance will be issued. A Release of Claims document must be executed and submitted to the Contracting Officer before final payment can occur.
- 11.5 Warranties
- A. Construction Warranty shall be based on date of final acceptance of the work and remain in effect for one year.
 - B. Provide all manufacturer warranty information of products installed organized by Division. All contact information, extent of coverage and conditions shall be clearly noted as well as date of overall final acceptance.
 - C. Information shall include operation and maintenance manuals for each warranted product.

ITEM 12 - DEFINITION OF CONTRACT LINE ITEMS

12.1 Summary

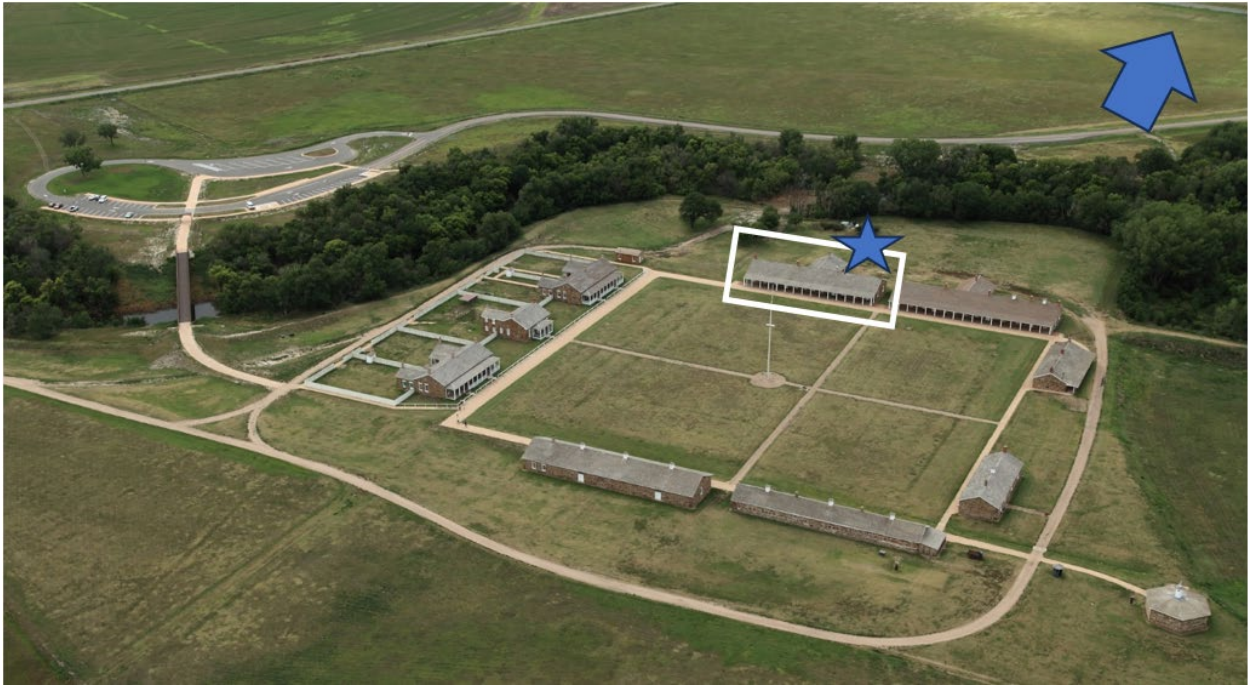
- A. The intent of this section is to explain, in general, what is and what is not included in a contract line item, and the limits or cut off points where one item ends, and another begins.
- B. If no contract line item exists for a portion of the work, include the costs in a related item.

12.2 List of Contract Line Items

- A. Contract Line-Item No. 010 Roof Surface Replacement of Visitor Center/Administration Building
 - 1. This item consists of Replace wood shingle roof in kind.
 - 2. Payment will be made per the contract and percentage of completion level.

END OF SECTION

Appendix A Site Plan



This photo is of the structure that is the focus of the work for this contract in relationship to the complex of the fort. It is sited on the north, northwest corner of the parade ground, within the historic corridor of Fort Larned NHS.

Appendix B Project Photos



Visitor Center/Administration building from southwest corner.



Northwest elevation of the building the focus of this contract scope



Northern elevation eastern half of the structure.



Southeast elevation detailing the transition of the two different slope of the roof.



Southern elevation of the visitor center building.

Appendix C Project Drawings

See attached Hover Report with all approximate dimensions and slopes.

SECTION 073129**WOOD SHINGLES****PART 1 - GENERAL****1.1 SCOPE OF WORK**

- A. Project Location: Fort Larned National Historic Site, 1767 Kansas State Highway 156 Larned, Kansas Northwest quadrant of the parade ground
- B. Building Name/Identifier: Visitor Center-Administration-Barracks (HS-1)

1.2 SUMMARY**A. Section Includes:****1. Selective Demolition**

- a. Remove the anchoring hardware of the lightening protection system from the roof surface but do-not disconnect lightening protection system. Lightening protection system is to be resecured to the new roof surface. If lightening protection system connections is broken, the contractor will have to have UL certification inspection performed.
- b. Remove approximately 10538 Square Feet (SF) of western red cedar shingles from roof. Remove approximately 15% of the skip-sheeting board that have deterioration. Including nails, do not set but remove shingle nails from sheeting boards.

2. Wood roof shingles.

- a. Install replacement sheeting matching dimensional measurement, wood species of the removed material and fastener type.
- b. Install approximately 10538 SF Grade 1 Class B fire retardant western red cedar shingles.

3. Underlayment materials. NA**4. Ridge vents.NA****5. Sheet metal flashing and trim.**

- a. Install Valley and step flashing on roof penetrations, reuse of existing apron and counter flashing is acceptable.

6. Gutter and downspouts. Reinstalled if removed. Gutters are 10-foot section centered over doorways.**7. Vents and plumbing boots. NA****8. Drainage mat.NA****1.3 PREINSTALLATION MEETINGS**

- A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

- A. Product Data: For the following:
 - 1. Wood roof shingles.
 - 2. Sheet metal flashing and trim.
- B. Shop Drawings: Metal Flashing
- C. Samples: For each exposed product.

1.5 INFORMATIONAL SUBMITTALS

- A. Sample warranty.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.7 SUBMITTAL COORDINATION

- A. Submittal List: A submittal list has been attached to the end of this Specification Section. The intent is to provide an overall summary of the submittal requirements and not a comprehensive list. The requirements of the individual Specification Sections, terms and conditions of the Contract still apply regardless of what is shown on the submittal list.

1.8 QUALITY ASSURANCE

- A. Installer Qualifications: Cedar Shake and Shingle Bureau (CSSB) member.
- B. Grading Agency Qualifications: An independent testing and inspecting agency recognized by authorities having jurisdiction as qualified to label wood products for compliance with referenced grading rules.

1.9 WARRANTY

- A. Materials Warranty: Manufacturer's warranty administered by CSSB and on CSSB's standard form in which manufacturer agrees to repair or replace CSSB-labeled products that fail in materials within specified warranty period. Material failures include manufacturing defects that result in leaks.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Exterior Fire-Test Exposure: Provide roofing materials identical to those of assemblies tested for fire resistance in accordance with American Society of Testing Materials (ASTM) E108 or UL 790 by Underwriters Laboratories or another testing and inspecting agency acceptable to authorities having jurisdiction. Identify products with appropriate markings of applicable testing agency.
 - 1. Wood Products: Class B.

- a. Fire-Retardant Treatment: Exterior-type pressure treatment complying with American Wood Protection Association (AWPA) U1 and AWPA T1.
 - b. Accelerated Weathering: Subject test specimens to ASTM D2898 Method A before testing.
 - c. Identification: Attach a label to each bundle of wood products; include identification mark of testing agency acceptable to authorities having jurisdiction and identify manufacturer, chemical treatment, method of application, purpose of treatment, and warranties available.
- B. Grading Rules: Provide wood products that comply with CSSB grading rules for products indicated.
1. Identification: Attach a label to each bundle of wood products that identifies manufacturer, type of product, grade, dimensions, and identification mark of grading agency acceptable to authorities having jurisdiction.

2.2 WOOD ROOF SHINGLES

- A. Cedar Shingles: Smooth-sawn western red cedar shingles.
1. Grade: No. 1, with starter courses of No. 1.
 2. Exposure:
 - a. Roof Slopes 3:12 to 4:12: 4-1/2 inches (110mm) maximum.
 - b. Roof Slopes 4:12 and Steeper: 5-1/2 inches (140mm) maximum.
 3. Size: 18 inches (457 mm) long; 0.45 inch (11 mm) thick at butt.
 4. Shingle Widths; Minimum 3-inches and Maximum 9-inches
- B. Cedar Ridge Board Units: Manufactured, smooth-sawn western red cedar premium grade 1-inch x 8-inch boards. Offset miters end joints, overlapping beveled long edge and weather seal underlayment self-adhering Polymer-Modified Bitumen Sheet ridge board cap alternately overlapped, and nailed.
1. Grade: No. 1.
 2. Length: 16-foot maximum length and 4-foot minimum as recommended for the roof application by the manufacturer.
 - 3.

2.3 UNDERLAYMENT MATERIALS NA

- A. Felt: Asphalt-saturated organic felts, nonperforated and complying with the following: NA
1. ASTM D226/D226M: Type II, No. 30.
 2. Use as underlayment.
- B. Valleys, Slope-Transitions, Eaves, Rake, and Roof/Wall Intersections: Install Self-Adhering, Polymer-Modified Bitumen Sheet: ASTM D1970/D1970M, minimum 55-mil- (1.4-mm-) thick sheet; glass-fiber-mat-reinforced, polymer-modified asphalt; with slip-resistant top surface and release backing; cold applied. Provide primer for adjoining concrete, masonry, and metal surfaces to receive underlayment. Approved product includes but is not limited to the following:
1. Grace Ice & Water Shield; Grace Construction Products.
 2. Winter Guard Waterproofing underlayment; CertainTeed.

3. Rhino Roof Granulated Self-Adhered underlayment.
4. Other approved equals.

2.4 VENTS NA

- A. Rigid Ridge Vent: NA
- B. Hood / Static Vent: NA
- C. Continuous Soffit Vents: NA
- D. Under-eave Vents: NA

2.5 ACCESSORIES

- A. Asphalt Roofing Cement: ASTM D4586/D4586M Type II, asbestos free.
- B. Roofing Nails: ASTM F1667, stainless steel, Type 304, or stainless steel, Type 316 box-type wire nails, sharp pointed, and of sufficient length to penetrate a minimum of 3/4 inch (19 mm) into sheathing or to penetrate through roof sheathing less than 3/4 inch (19 mm) thick.
 1. Where nails are in contact with metal flashing, use nails made from same metal as flashing.
- C. Asphalt-Saturated Organic Felt Underlayment Fasteners: NA.

2.6 SELECTIVE DEMOLITION

- A. General: Demolish and remove existing construction only to the extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:
 1. Full Roof Tear-off: Removal of existing roofing system down to existing roof deck or structure.
 2. Temporarily protect existing roofing system and components that are to remain as required from damage.
 3. Existing roof drains and roof drainage shall be protected during each stage of reroofing, and roof-drain plugs shall be installed when demolition is taking place so that debris cannot be lost down the drains. Contractor shall remove all drain plugs when demolition is completed in the area affecting the drain.
 4. Do not remove more roof than can be replaced in a day, and the roof will be made watertight each night.
 5. Remove debris and materials so as not to impose excessive loads on supporting roof, structure, or framing.
 6. Remove debris and material that falls into attic space document attic area prior to tear-off.
 7. Dispose of demolished items and materials promptly.
- B. Removed and Reinstalled Items:
 1. Clean and repair items to functional condition adequate for intended reuse.
 2. Reinstall items in locations indicated, or in original locations as required. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.
 3. Lightning protection to be removed and reinstalled by a certified installer.

- C. Existing Items to Remain: Protect construction indicated to remain against damage and soiling during selective demolition. Protect antennas, Valley Flashing.
- D. Inspect deck after tear-off of roofing system.
 - 1. If broken or loose fasteners that secure deck panels to one another or to structure are observed, or if deck appears or feels inadequately attached, immediately notify COR.
 - a. Do not proceed with installation until directed by COR.
 - 2. If deck surface is unsuitable for receiving new roofing or if structural integrity of deck is suspect, immediately notify COR.
 - a. Do not proceed with installation until directed by COR.
- E. Disposal of Demolished Materials:
 - 1. General: Except for items or materials indicated to be reused, salvaged, reinstalled, or otherwise indicated to remain NPS's property, remove demolished materials from Project site.
 - a. Do not allow demolished materials to accumulate on-site.
 - b. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
 - c. Cover any accumulated demolition material in such manner to prevent displacement from wind, including material in waste containers.
 - 2. Burning: Do not burn demolished materials.
 - 3. Disposal: Transport demolished materials off NPS's property and legally dispose of them.
- F. Cleaning:
 - 1. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

2.7 SHEET METAL FLASHING AND TRIM

- A. Performance Requirements:
 - 1. General: Sheet metal flashing and trim assemblies shall withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim shall not rattle, leak, or loosen, and shall remain watertight.
 - 2. Sheet Metal Standard for Flashing and Trim: Comply with National Roofing Contractors Association's (NRCA's) "The NRCA Roofing Manual" and Sheet Metal and Air Conditioning Contractors National Association's (SMACNA's) "Architectural Sheet Metal Manual" requirements for dimensions and profiles shown unless more stringent requirements are indicated.
 - 3. Single Ply Roofing Industry (SPRI) Wind Design Standard: Manufacture and install copings and roof edge flashings tested according to SPRI ES-1 and capable of resisting the following design pressure:

- a. Design Pressure: 30-psf vertical design pressure.
 4. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
- B. Sheet Metal:
1. General: Protect mechanical and other finishes on exposed surfaces from damage by applying strippable, temporary protective film before shipping.
 2. Metallic-Coated Steel Sheet: Provide zinc-plated 16oz Cooper steel sheet according to ASTM A 653/A 653M, G90 coating designation or aluminum-zinc alloy-coated steel sheet according to ASTM A 792/A 792M, Class AZ50 coating designation, Grade 40; repainted by coil-coating process to comply with ASTM A 755/A 755M.
 - a. Surface: Smooth, flat.
 - b. Exposed Coil-Coated Finish:
 - 1) Two-Coat Fluoropolymer. American Architectural Manufacturers Association (AAMA) 621. Fluoropolymer finish containing not less than 70 percent Polyvinylidene Fluoride (PVDF) resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
 - c. Color: As selected by COR from manufacturer's full range.
 - d. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester backer finish, consisting of prime coat and wash coat with minimum total dry film thickness of 0.5 mil.
- C. Miscellaneous Materials:
1. General: Provide materials and types of fasteners, solder, protective coatings, sealants, and other miscellaneous items as required for complete sheet metal flashing and trim installation and as recommended by manufacturer of primary sheet metal or manufactured item unless otherwise indicated.
 2. Fasteners: Wood screws, annular threaded nails, self-tapping screws, self-locking rivets and bolts, and other suitable fasteners designed to withstand design loads and recommended by manufacturer of primary sheet metal or manufactured item.
 - a. General: Blind fasteners or self-drilling screws, gasketed, with hex-washer head.
 - 1) Exposed Fasteners: Heads matching color of sheet metal using plastic caps or factory-applied coating. Provide metal-backed Ethylene Propylene Diene Monomer (EPDM) or Polyvinyl Chloride (PVC) sealing washers under heads of exposed fasteners bearing on weather side of metal.
 - 2) Blind Fasteners: High-strength aluminum or stainless-steel rivets suitable for metal being fastened.
 - b. Fasteners for Zinc-Coated (Galvanized) or Aluminum-Zinc Alloy-Coated Steel Sheet: Series 300 stainless steel or hot-dip galvanized steel according to ASTM A 153/A 153M or ASTM F 2329.

3. Sealant Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, non-sag, nontoxic, nonstaining tape 13 mm (1/2 inch) wide and 3 mm (1/8 inch) thick.
4. Elastomeric Sealant: ASTM C 920, elastomeric polymer sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain watertight.
5. Butyl Sealant: ASTM C 1311, single-component, solvent-release butyl rubber sealant; polyisobutylene plasticized; heavy bodied for hooked-type expansion joints with limited movement.

D. Fabrication General:

1. General: Custom fabricate sheet metal flashing and trim to comply with details shown and recommendations in cited sheet metal standard that apply to design, dimensions, geometry, metal thickness, and other characteristics of item required. Fabricate sheet metal flashing and trim in shop to greatest extent possible.
 - a. Fabricate sheet metal flashing and trim in thickness or weight needed to comply with performance requirements, but not less than that specified for each application and metal.
 - b. Obtain field measurements for accurate fit before shop fabrication.
 - c. Form sheet metal flashing and trim to fit substrates without excessive oil canning, buckling, and tool marks; true to line, levels, and slopes; and with exposed edges folded back to form hems.
 - d. Conceal fasteners and expansion provisions where possible. Do not use exposed fasteners on faces exposed to view.
2. Fabrication Tolerances: Fabricate sheet metal flashing and trim that is capable of installation to a tolerance of 1/4 inch in 20 feet on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.
3. Expansion Provisions: Form metal for thermal expansion of exposed flashing and trim.
 - a. Form expansion joints of intermeshing hooked flanges, not less than 25 mm (1 inch) deep, filled with butyl sealant concealed within joints.
 - b. Use lapped expansion joints only where indicated on Drawings.
4. Sealant Joints: Where movable, non-expansion-type joints are required, form metal to provide for proper installation of elastomeric sealant according to cited sheet metal standard.
5. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal.
6. Fabricate cleats and attachment devices of sizes as recommended by cited sheet metal standard for application, but not less than thickness of metal being secured.
7. Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with elastomeric sealant unless otherwise recommended by sealant manufacturer for intended use.

E. Roof-Drainage Sheet Metal Fabrications: NA

F. Miscellaneous Roof Sheet Metal Fabrications:

1. Roof-Edge Flashing and Cap Flashing: NA

2. Copings: NA
3. Base Flashing: Fabricate from the following materials:
 - a. zinc-coated (galvanized) steel sheet according to ASTM A 653/A 653M, G90 coating designation 0.024.
4. Counterflashing and Flashing Receivers: Fabricate from the following materials:
 - a. zinc-coated (galvanized) steel sheet according to ASTM A 653/A 653M, G90 coating designation 0.024.
5. Roof-Penetration Flashing: Fabricate from the following materials:
 - a. zinc-coated (galvanized) steel sheet according to ASTM A 653/A 653M, G90 coating designation 0.024.
6. Vent-Pipe Flashings: NA

PART 3 - EXECUTION

3.1 INSTALLATION OF UNDERLAYMENT MATERIALS

- A. Comply with underlayment manufacturer's written installation instructions and with recommendations in CSSB's "New Roof Construction Manual" and National Roofing Contractors Association's (NRCA's) "The NRCA Roofing Manual: Steep-Slope Roof Systems" applicable to products and applications indicated unless more stringent requirements are specified in the Section or indicated on Drawings.
- B. Felt: NA
- C. Install Self-Adhering, Polymer-Modified Bitumen Sheet: Install, wrinkle free, on roof deck in locations indicated on Drawings or SOW. Install self-adhering polymer-modified bitumen sheet under ridge board cap.
 1. Comply with low-temperature installation restrictions of underlayment manufacturer.
 2. Install lapped in direction that sheds water.
 3. Lap sides not less than 4 inches (102 mm). Lap ends not less than 6 inches (152 mm), staggered 24 inches (610 mm) between succeeding courses.
 4. Roll laps with roller.
 5. Prime concrete, masonry, and metal surfaces to receive self-adhering sheet if required in writing by the manufacturer for installation of their underlayment.
 6. Cover underlayment within seven days.
 7. Solidly cement valley underlayment to roof-field underlayment that is woven through valley using asphalt roofing cement.

3.2 INSTALLATION OF METAL FLASHING AND TRIM

- A. Examination:
 1. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, substrate, and other conditions affecting performance of the Work.

- a. Verify compliance with requirements for installation tolerances of substrates.
 - b. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
 - c. Verify that air- or water-resistant barriers have been installed over sheathing or backing substrate to prevent air infiltration or water penetration.
2. Proceed with installation only after unsatisfactory conditions have been corrected.

B. General Installation:

1. General: Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement. Use fasteners, solder, protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.
 - a. Install sheet metal flashing and trim true to line, levels, and slopes. Provide uniform, neat seams with minimum exposure of solder, welds, and sealant.
 - b. Install sheet metal flashing and trim to fit substrates and to result in watertight performance. Verify shapes and dimensions of surfaces to be covered before fabricating sheet metal.
 - c. Space cleats not more than 300 mm (12 inches) apart. Attach each cleat with at least two fasteners. Bend tabs over fasteners.
 - d. Install exposed sheet metal flashing and trim with limited oil canning, and free of buckling and tool marks.
 - e. Torch cutting of sheet metal flashing and trim is not permitted.
 - f. Do not use graphite pencils to mark metal surfaces.
2. Metal Protection: Where dissimilar metals contact each other, or where metal contacts pressure-treated wood or other corrosive substrates, protect against galvanic action or corrosion by painting contact surfaces with bituminous coating or by other permanent separation as recommended by sheet metal manufacturer or cited sheet metal standard.
 - a. Coat concealed side of sheet metal flashing and trim with bituminous coating where flashing and trim contact wood, ferrous metal, or cementitious construction.
 - b. Underlayment: Where installing sheet metal flashing and trim directly on Cementitious or wood substrates, install underlayment and cover with slip sheet.
3. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim. Space movement joints at maximum of 3 m (10 feet) with no joints within 600 mm (24 inches) of corner or intersection.
 - a. Form expansion joints of intermeshing hooked flanges, not less than 25 mm (1 inch) deep, filled with sealant concealed within joints.
 - b. Use lapped expansion joints only where indicated on Drawings.
4. Fasteners: Use fastener sizes that penetrate substrate not less than recommended by fastener manufacturer to achieve maximum pull-out resistance.
5. Conceal fasteners and expansion provisions where possible in exposed work and locate to minimize possibility of leakage. Cover and seal fasteners and anchors as required for a tight installation.
6. Seal joints as required for watertight construction.
 - a. Embed hooked flanges of joint members not less than 1 inch into sealant.

- b. Form joints to completely conceal sealant.
 - c. When ambient temperature at time of installation is between 40 and 70 deg F, set joint members for 50 percent movement each way. Adjust setting proportionately for installation at higher ambient temperatures.
 - d. Do not install sealant-type joints at temperatures below 40 deg F.
 - C. Roof-Drainage System Installation: NA
 - D. Roof Flashing Installation:
 - 1. General: Install sheet metal flashing and trim to comply with performance requirements, sheet metal manufacturer's written installation instructions, and cited sheet metal standard. Provide concealed fasteners where possible, and set units true to line, levels, and slopes. Install work with laps, joints, and seams that are permanently watertight and weather resistant.
 - 2. Roof Edge Flashing: NA
 - 3. Copings: NA
 - 4. Pipe or Post Counterflashing: NA
 - 5. Counterflashing: Coordinate installation of counterflashing with installation of base flashing. Insert counterflashing in reglets or receivers and fit tightly to base flashing. Extend counterflashing 100 mm (4 inches) over base flashing. Lap counterflashing joints minimum of 100 mm (4 inches). Secure in waterproof manner by means of snap-in installation and sealant or lead wedges and sealant unless otherwise notes.
 - a. Roof-Penetration Flashing: Coordinate installation of roof-penetration flashing with installation of roofing and other items penetrating roof. Seal with elastomeric or butyl sealant and clamp flashing to pipes that penetrate roof per manufacturer's recommendations.
 - 6. Pipe Flashings: NA
 - E. Cleaning and Protection:
 - 1. Clean exposed metal surfaces of substances that interfere with uniform oxidation and weathering.
 - 2. Clean off excess sealants.
 - 3. Remove temporary protective coverings and strippable films as sheet metal flashing and trim are installed unless otherwise indicated in manufacturers written installation instructions. On completion of sheet metal flashing and trim installation, remove unused materials and clean finished surfaces as recommended by sheet metal flashing and trim manufacturer. Maintain sheet metal flashing and trim in clean condition during construction.
 - F. Replace sheet metal flashing and trim that have been damaged or that have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.
- 3.3 INSTALLATION OF WOOD ROOF SHINGLES
- A. General: Install wood roof shingles in accordance with manufacturer's written instructions and recommendations in CSSB's "New Roof Construction Manual" and NRCA's "The NRCA Roofing Manual: Steep-Slope Roofing Systems."

- B. Install wood-shingle starter course along lowest roof edge.
 - 1. Install in single layer.
 - 2. Extend 1 inch (25 mm) over fascia.
 - 3. Extend 1 inch (25 mm) over rake edge.
- C. Install first course of wood roof shingles directly over starter course and in continuous straight-line courses across roof deck. Install second and succeeding courses of wood roof shingles in continuous straight-line courses across roof deck.
 - 1. Extend 1 inch (25 mm) over rake edge.
 - 2. Offset joints between shingles in succeeding courses a minimum of 1-1/2 inches (38 mm). Do not align vertical joints within 3 alternate courses.
 - 3. Space shingles a minimum of 1/4 inch (6 mm) and a maximum of 3/8 inch (10 mm) apart.
 - 4. Fasten each shingle with two nails spaced 3/4 to 1 inch (19 to 25 mm) from edge of shingle and 1-1/2 to 2 inches (38 to 51 mm) above butt line of succeeding course. Drive fasteners flush with top surface of shingles without crushing wood. Increase as required to meet all criteria listed herein.
- D. Open Valleys: Cut and fit wood roof shingles at open valleys, trimming upper concealed corners of shingles. Maintain uniform width of exposed open valley from highest to lowest point.
- E. Ridge Vents: NA
- F. Ridge Units: Install units over wood roof shingles trimmed at apex of ridges and hips.
 - 1. At unventilated ridges and hips, install concealed strip of self-adhering, polymer-modified bitumen sheet underlayment over apex shingles and below ridge units.

END OF SECTION

SUBMITTAL LIST

Note: See tab below for Samples and Instructions. Attach to end of Specification Section 07 31 29 when complete.

[illegible]

**VIEW 3D MODEL**

Areas	Siding	Other
Facades	14 ft ²	4469 ft ²
Openings	-	591 ft ²
Trims*	40 ft ²	786 ft ²
Unknown (no photos)*	-	-
Total	54 ft ²	5846 ft ²

*Any trim or unknown material that touches siding is included in the 'Siding' column. If it does not touch siding, then it's included in the 'Other' column.

Openings	Siding	Other
Quantity	0	34
Tops Length	-	107' 8"
Sills Length	-	97' 3"
Sides Length	-	388' 10"
Total Perimeter	-	593' 10"

Corners	Siding	Other
Inside Qty	0	2
Inside Length	-	19'
Outside Qty	0	26
Outside Length	-	169' 10"

Accessories	Siding	Other
Shutter Qty	0	0
Shutter Area	0 ft ²	0 ft ²
Vents Qty	0	0
Vents Area	0 ft ²	0 ft ²

Trim	Siding	Other
Level Starter	17' 9"	836' 7"
Sloped Trim	-	32' 11"
Vertical Trim	2' 5"	162' 5"

Roofline	Length	Avg. Depth	Soffit Area
Eaves Fascia	330' 5"	-	-
Level Frieze Board	332'	3'	2164 ft ²
Rakes Fascia	164' 4"	-	-
Sloped Frieze Board	157' 5"	1' 1"	198 ft ²

SIDING WASTE TOTALS

Siding & Trim Only*	Area	Squares
Zero Waste	26 ft ²	0½
+10%	29 ft ²	0½
+18%	30 ft ²	0½

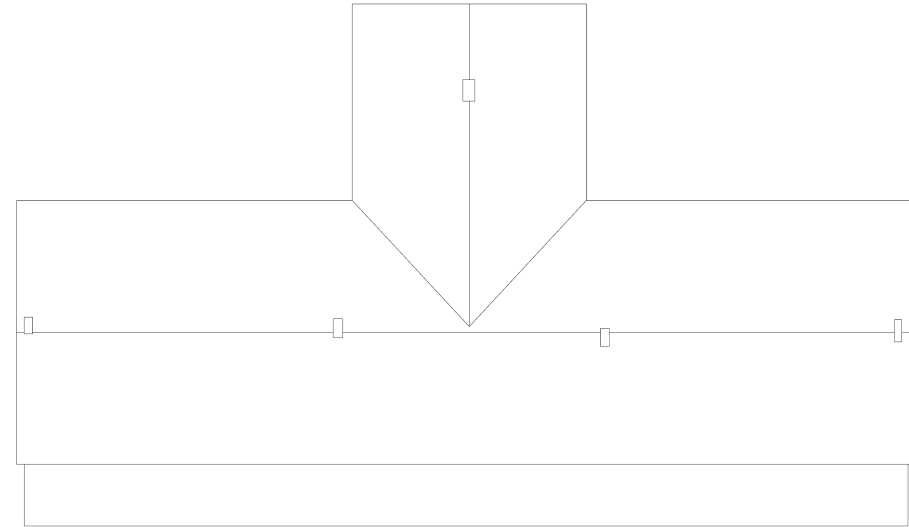
+ Openings < 20ft ²	Area	Squares
Zero Waste	26 ft ²	0½
+10%	29 ft ²	0½
+18%	30 ft ²	0½

+ Openings < 33ft ²	Area	Squares
Zero Waste	26 ft ²	0½
+10%	29 ft ²	0½
+18%	30 ft ²	0½

*The first three rows of the Siding Waste Factor table are calculated using the total ft² of siding facades, ft² of trim touching siding, and ft² of unknowns touching siding.

Roof	Area	Total	Length
Roof Facets	10538 ft ²	5	-
Ridges / Hips	-	7	197' 1"
Valleys	-	2	61' 8"
Rakes	-	8	164' 4"
Eaves	-	7	330' 5"
Flashing	-	10	18' 9"
Step Flashing	-	20	32' 11"
Drip Edge/Perimeter	-	-	494' 9"

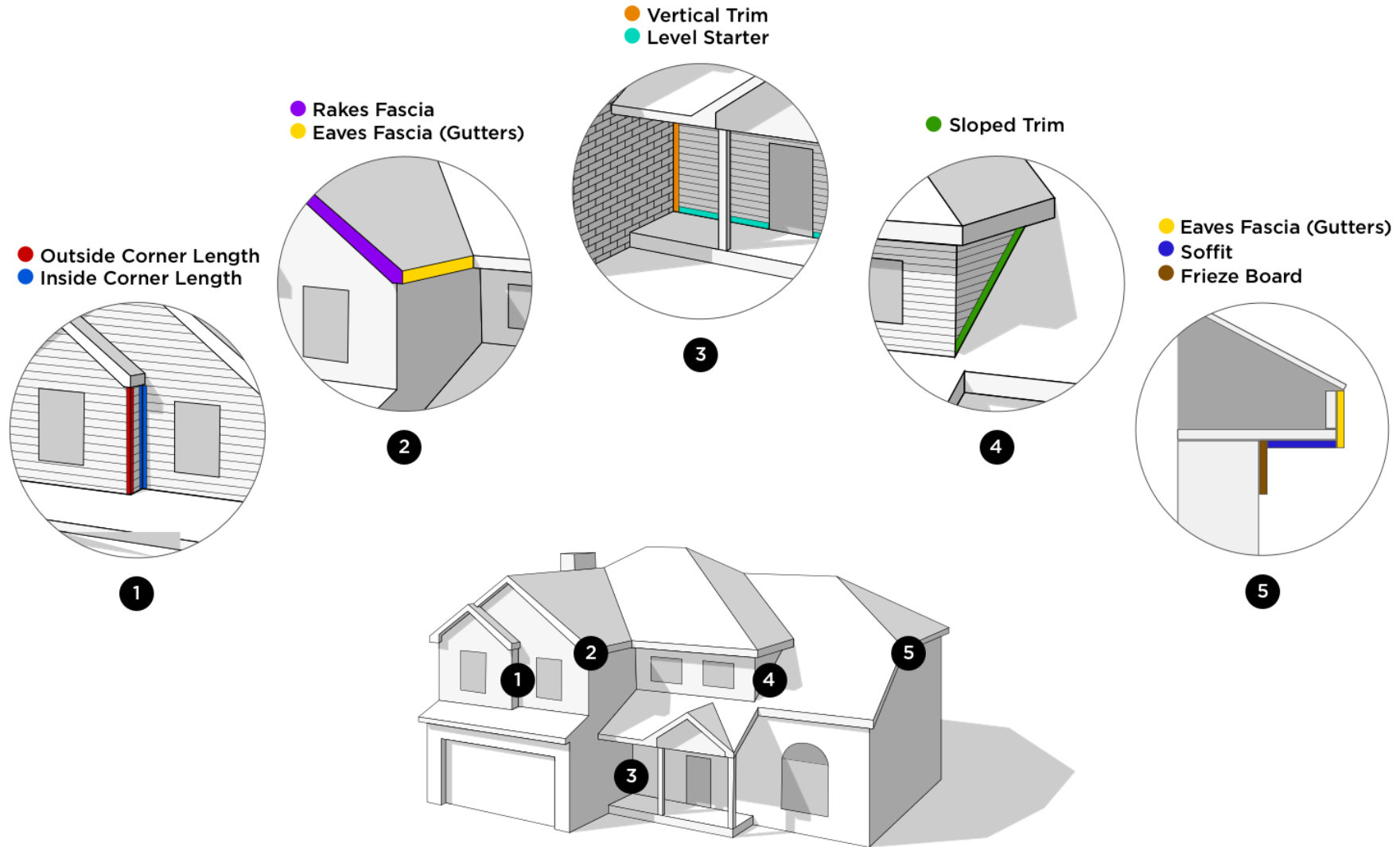
Roof Pitch*	Area	Percentage
6 / 12	8945 ft ²	84.88%
2 / 12	1593 ft ²	15.12%

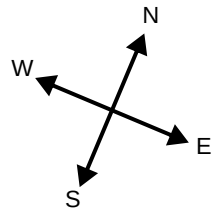


Example Waste Factor Calculations

	Zero Waste	+5%	+10%	+15%	+20%
Area	10538 ft ²	11065 ft ²	11592 ft ²	12119 ft ²	12646 ft ²
Squares	105⅔	110⅔	116	121⅓	126⅔

The table above provides the total roof area of a given property using waste percentages as noted. Please consider that area values and specific waste factors can be influenced by the size and complexity of the property, captured image quality, specific roofing techniques, and your own level of expertise. Additional square footage for Hip, Ridge, and Starter shingles are not included in this waste factor and will require additional materials. This table is only intended to make common waste calculations easier and should not be interpreted as recommendations.

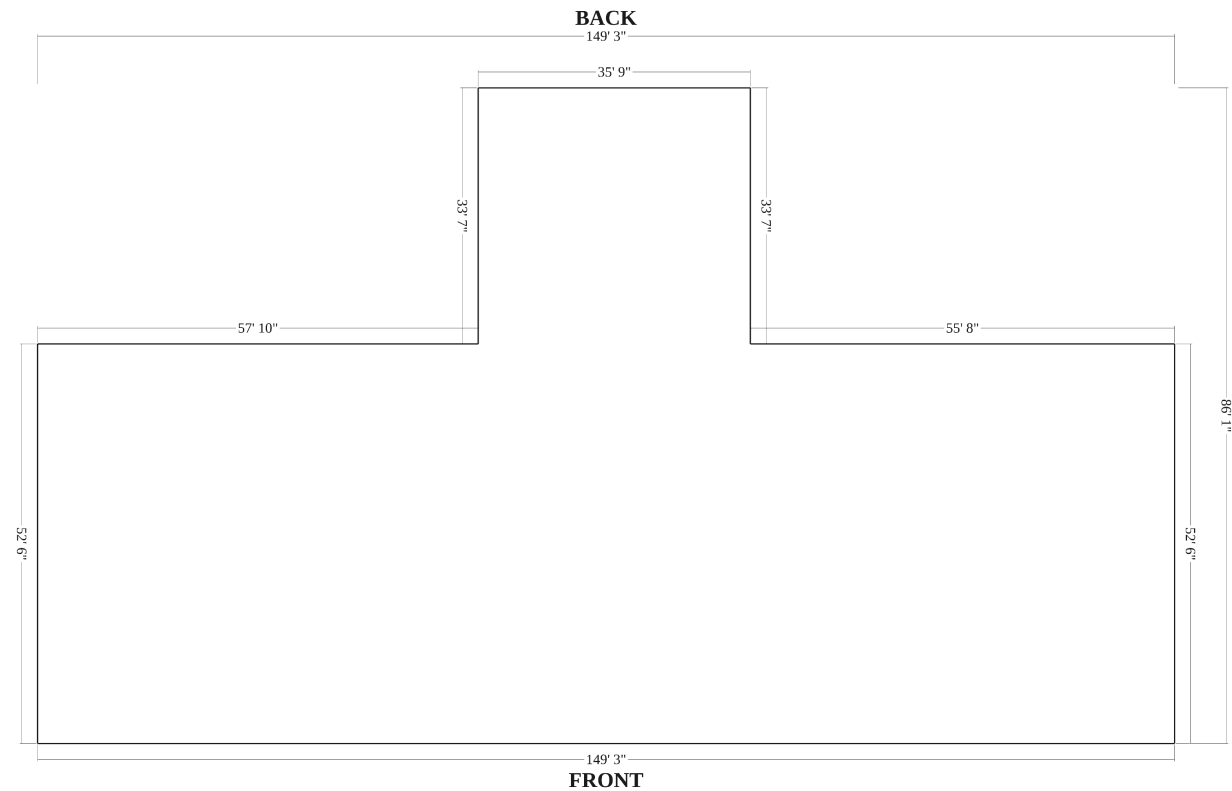




Number of Stories: 1

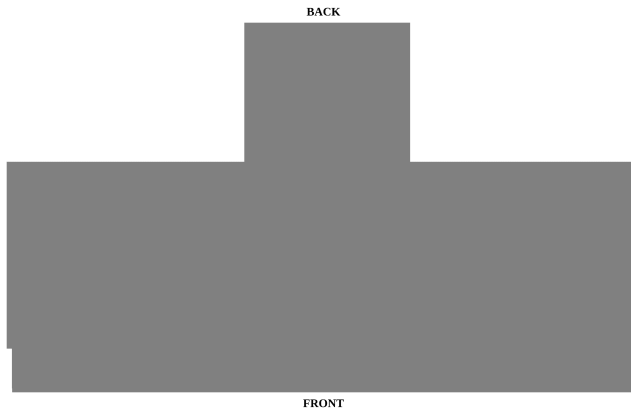
Footprint Perimeter: 470' 8"

Footprint Area: 9032 ft²

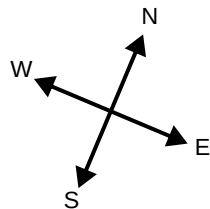


Siding Per Elevation

FRONT	RIGHT			LEFT			BACK
	SI-1	-	7 ft ²	SI-2	-	7 ft ²	
0 ft²	7 ft²			7 ft²			0 ft²

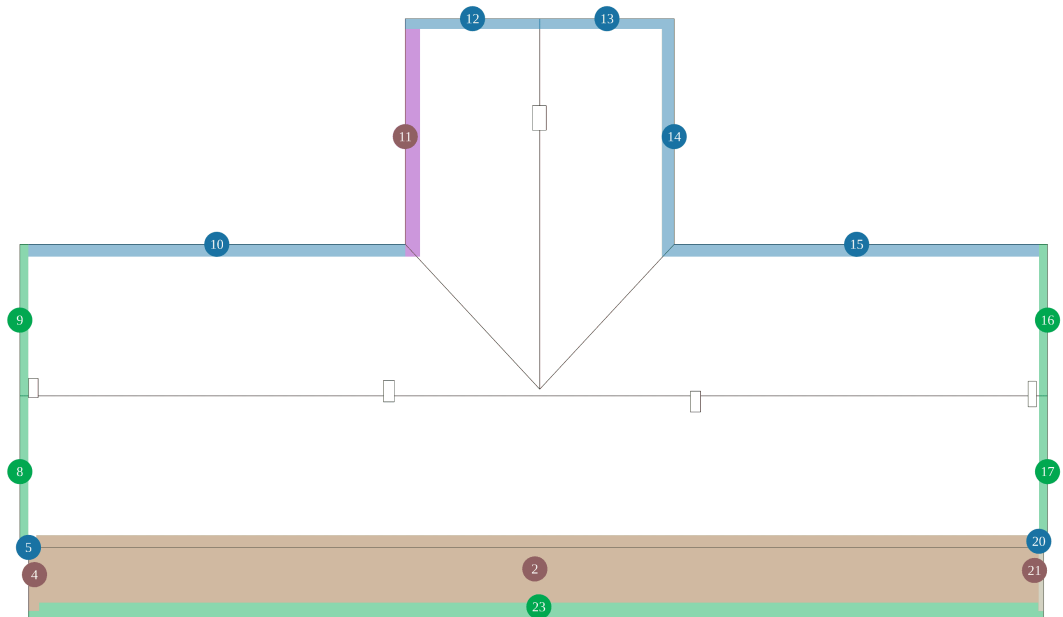


Number of Stories: 1



Soffit Summary

Depth	Type	Count	Total Length	Total Area
1" - 6"	rakes	1	9' 6"	2 ft ²
	eaves	4	11"	0 ft ²
6" - 12"	rakes	1	9' 6"	7 ft ²
12" - 18"	rakes	4	98' 8"	123 ft ²
	eaves	2	297' 5"	347 ft ²
18" - 24"	rakes	2	44' 5"	67 ft ²
	eaves	5	143' 6"	259 ft ²
24" - 48"	eaves	1	33' 7"	73 ft ²
> 48"	eaves	3	149'	1484 ft ²
Totals			786' 7"	2362 ft ²

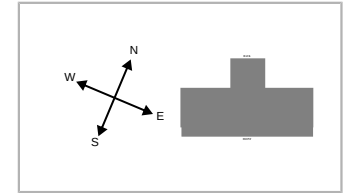


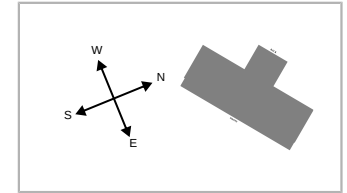
Soffit Breakdown

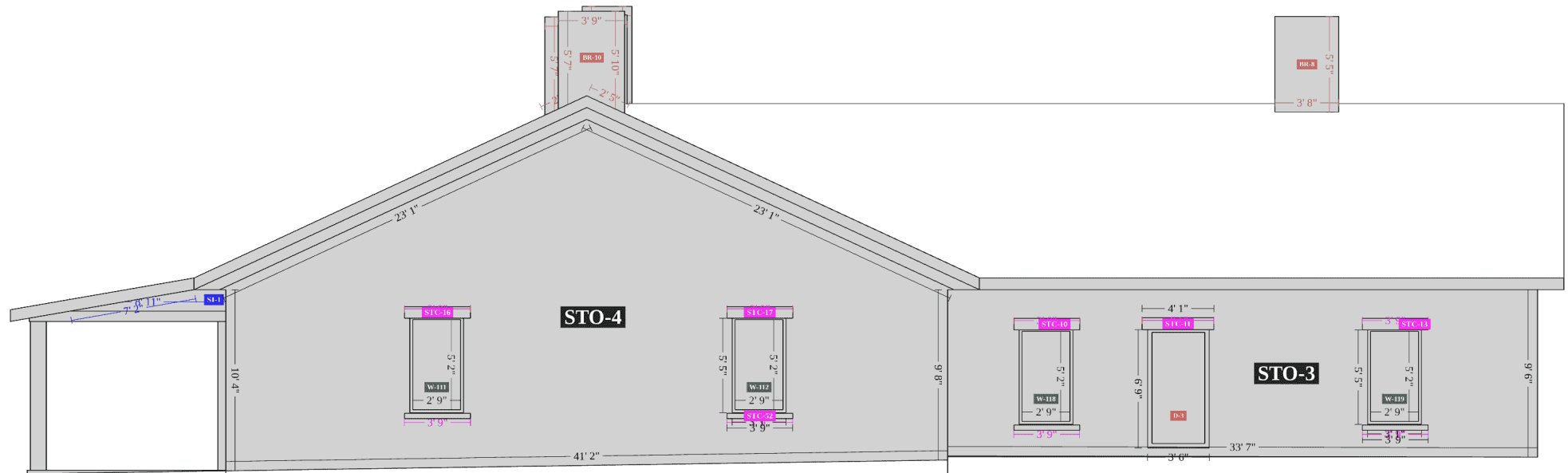
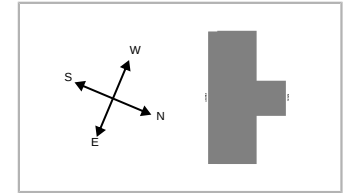
	num	Type	Depth	Length	Area	Pitch
Ø	1	eave	13"	149' 11"	164 ft ²	2 / 12
●	2	eave	120"	146' 5"	1458 ft ²	2 / 12
Ø	3	rake	2"	9' 6"	2 ft ²	2 / 12
●	4	eave	129"	1' 5"	15 ft ²	2 / 12
●	5	eave	22"	3"	0 ft ²	6 / 12
Ø	6	eave	5"	5"	0 ft ²	6 / 12
Ø	7	eave	5"	1"	0 ft ²	6 / 12
●	8	rake	15"	24' 8"	31 ft ²	6 / 12
●	9	rake	15"	24' 8"	31 ft ²	6 / 12
●	10	eave	22"	55' 8"	101 ft ²	6 / 12
●	11	eave	26"	33' 7"	73 ft ²	6 / 12
●	12	rake	18"	22' 2"	33 ft ²	6 / 12
●	13	rake	18"	22' 2"	33 ft ²	6 / 12
●	14	eave	22"	33' 7"	61 ft ²	6 / 12
●	15	eave	22"	53' 11"	97 ft ²	6 / 12
●	16	rake	15"	24' 8"	31 ft ²	6 / 12
●	17	rake	15"	24' 8"	31 ft ²	6 / 12
Ø	18	eave	5"	1"	0 ft ²	6 / 12
Ø	19	eave	5"	4"	0 ft ²	6 / 12

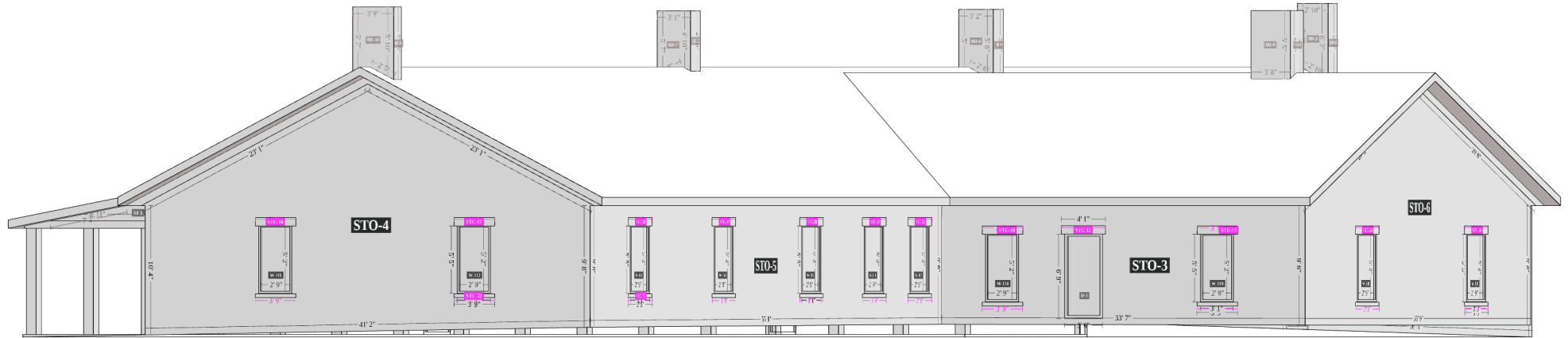
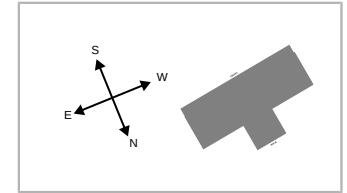
Ø Feature is too small to label on the plan diagram

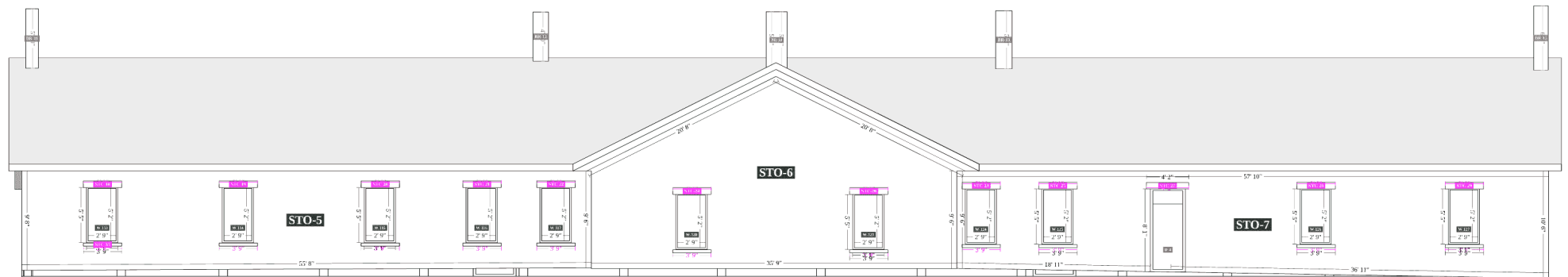
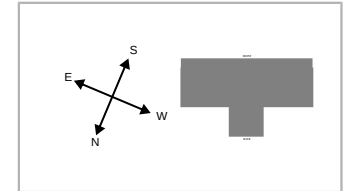
	num	Type	Depth	Length	Area	Pitch
●	20	eave	22"	1"	0 ft ²	6 / 12
●	21	eave	114"	1' 2"	11 ft ²	2 / 12
Ø	22	rake	9"	9' 6"	7 ft ²	2 / 12
●	23	eave	15"	147' 7"	183 ft ²	2 / 12

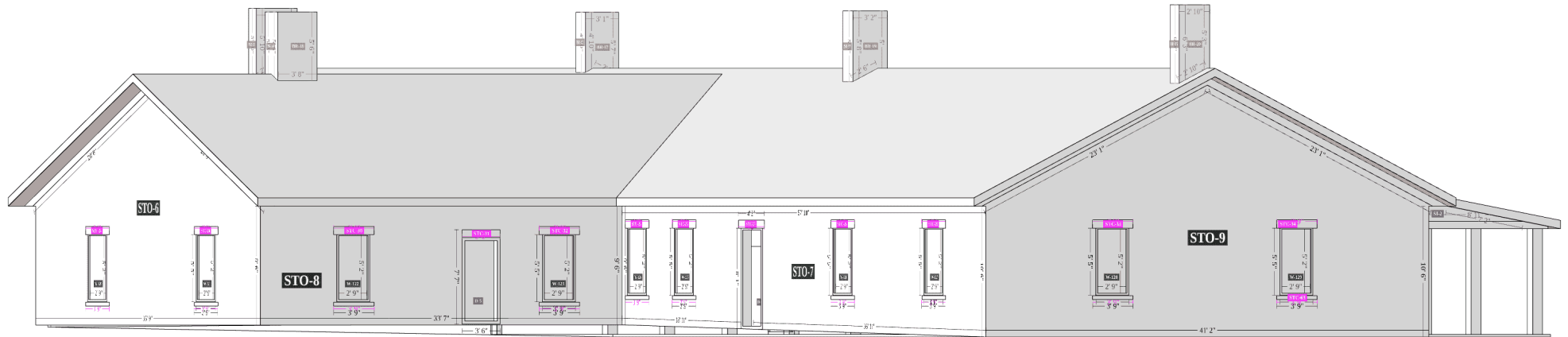
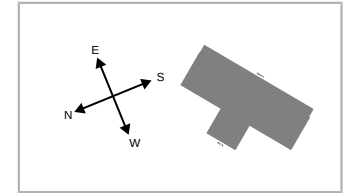


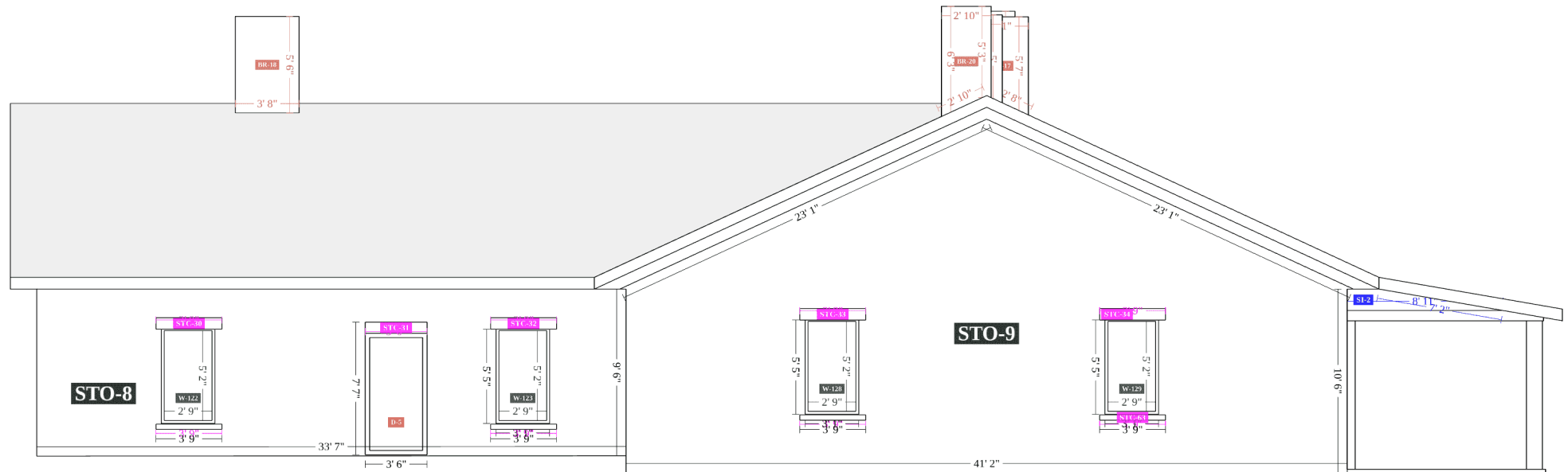
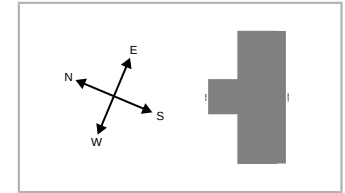


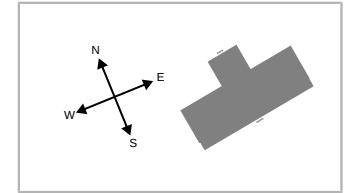












Siding

Facade	Area	Inside Corners	Outside Corners	Openings	Shutters	Vents
SI-1	7 ft ²	-	-	-	-	-
SI-2	7 ft ²	-	-	-	-	-
Total	14 ft ²	0	0	0	0	0

Brick

Facade	Area	Openings	Shutters	Vents
BR-1	7 ft ²	-	-	-
BR-2	16 ft ²	-	-	-
BR-3	8 ft ²	-	-	-
BR-4	16 ft ²	-	-	-
BR-5	8 ft ²	-	-	-
BR-6	11 ft ²	-	-	-
BR-7	15 ft ²	-	-	-
BR-8	20 ft ²	-	-	-
BR-9	7 ft ²	-	-	-
BR-10	20 ft ²	-	-	-
BR-11	7 ft ²	-	-	-
BR-12	7 ft ²	-	-	-
BR-13	9 ft ²	-	-	-
BR-14	11 ft ²	-	-	-
BR-15	9 ft ²	-	-	-
BR-16	20 ft ²	-	-	-
BR-17	15 ft ²	-	-	-
BR-18	20 ft ²	-	-	-
BR-19	16 ft ²	-	-	-

Brick (cont.)

Facade	Area	Openings	Shutters	Vents
BR-20	16 ft ²	-	-	-
Total	258 ft ²	0	0	0

Stone

Facade	Area	Openings	Shutters	Vents
STO-1	3 ft ²	-	-	-
STO-2	1059 ft ²	2	-	-
STO-3	253 ft ²	-	-	-
STO-4	575 ft ²	-	-	-
STO-5	434 ft ²	-	-	-
STO-6	472 ft ²	-	-	-
STO-7	468 ft ²	1	-	-
STO-8	253 ft ²	-	-	-
STO-9	594 ft ²	-	-	-
STO-10*	1 ft ²	-	-	-
Total	4112 ft ²	3	0	0

* Facet is not visible due to size or location

Stucco

Facade	Area	Openings	Shutters	Vents
STC-1	2 ft ²	1	-	-
STC-2	2 ft ²	1	-	-
STC-3	2 ft ²	1	-	-
STC-4	2 ft ²	-	-	-
STC-5	2 ft ²	1	-	-
STC-6	2 ft ²	1	-	-
STC-7	2 ft ²	1	-	-
STC-8	2 ft ²	1	-	-
STC-9	2 ft ²	-	-	-
STC-10	2 ft ²	1	-	-
STC-11	3 ft ²	1	-	-
STC-12	2 ft ²	1	-	-
STC-13	2 ft ²	1	-	-
STC-14	2 ft ²	1	-	-
STC-15	2 ft ²	1	-	-
STC-16	2 ft ²	1	-	-
STC-17	2 ft ²	1	-	-
STC-18	2 ft ²	1	-	-
STC-19	2 ft ²	1	-	-

Stucco (cont.)

Facade	Area	Openings	Shutters	Vents
STC-20	2 ft ²	1	-	-
STC-21	2 ft ²	1	-	-
STC-22	2 ft ²	1	-	-
STC-23	2 ft ²	1	-	-
STC-24	2 ft ²	1	-	-
STC-25	2 ft ²	1	-	-
STC-26	2 ft ²	1	-	-
STC-27	3 ft ²	-	-	-
STC-28	2 ft ²	1	-	-
STC-29	2 ft ²	1	-	-
STC-30	2 ft ²	1	-	-
STC-31	2 ft ²	1	-	-
STC-32	2 ft ²	1	-	-
STC-33	2 ft ²	1	-	-
STC-34	2 ft ²	1	-	-
STC-35*	1 ft ²	-	-	-
STC-36*	1 ft ²	-	-	-
STC-37*	1 ft ²	-	-	-
STC-38*	1 ft ²	-	-	-

Stucco (cont.)

Facade	Area	Openings	Shutters	Vents
STC-39*	1 ft ²	-	-	-
STC-40*	1 ft ²	-	-	-
STC-41*	1 ft ²	-	-	-
STC-42*	1 ft ²	-	-	-
STC-43*	1 ft ²	-	-	-
STC-44*	1 ft ²	-	-	-
STC-45*	1 ft ²	-	-	-
STC-46*	1 ft ²	-	-	-
STC-47*	1 ft ²	-	-	-
STC-48*	1 ft ²	-	-	-
STC-49*	1 ft ²	-	-	-
STC-50*	1 ft ²	-	-	-
STC-51*	1 ft ²	-	-	-
STC-52*	1 ft ²	-	-	-
STC-53*	1 ft ²	-	-	-
STC-54*	1 ft ²	-	-	-
STC-55*	1 ft ²	-	-	-
STC-56*	1 ft ²	-	-	-
STC-57*	1 ft ²	-	-	-

Stucco (cont.)

Facade	Area	Openings	Shutters	Vents
STC-58*	1 ft ²	-	-	-
STC-59*	1 ft ²	-	-	-
STC-60*	1 ft ²	-	-	-
STC-61*	1 ft ²	-	-	-
STC-62*	1 ft ²	-	-	-
STC-63*	1 ft ²	-	-	-
Total	99 ft ²	31	0	0

* Facet is not visible due to size or location

Facades

		Trim			Corners		Roofline		Openings		
Facade	Area	Level Starter	Sloped	Vertical	Inside	Outside	Level Frieze Board	Sloped Frieze Board	Tops	Sills	Sides
SI-1	7 ft ²	8' 11"	-	1' 3"	-	-	1' 10"	7' 2"	-	-	-
SI-2	7 ft ²	8' 11"	-	1' 3"	-	-	1' 10"	7' 2"	-	-	-
Total*	14 ft ²	17' 9"	-	2' 5"	-	-	3' 7"	14' 4"	-	-	-

*Totals de-duplicate any line segments that are shared between multiple facades, and as a result may not represent a total summation of the corresponding column.

Example Waste Factor Calculations

SIDING & TRIM ONLY

	Zero Waste	+10%	+18%
SI-1	7 ft ²	8 ft ²	8 ft ²
SI-2	7 ft ²	8 ft ²	8 ft ²
Trims	12 ft ²	13 ft ²	14 ft ²
Total	26 ft ²	29 ft ²	30 ft ²

+ OPENINGS < 20FT²

Zero Waste	+10%	+18%
7 ft ²	8 ft ²	8 ft ²
7 ft ²	8 ft ²	8 ft ²
12 ft ²	13 ft ²	14 ft ²
26 ft ²	29 ft ²	30 ft ²

+ OPENINGS < 33FT²

Zero Waste	+10%	+18%
7 ft ²	8 ft ²	8 ft ²
7 ft ²	8 ft ²	8 ft ²
12 ft ²	13 ft ²	14 ft ²
26 ft ²	29 ft ²	30 ft ²

The first Siding Waste Factor table is calculated using the total ft² of siding facades, ft² of trim touching siding, and ft² of unknowns touching siding.

The tables above provide the area of siding on a given property, segmented by individual and in sum total form. Values include openings (doors & windows) and waste percentages as noted. Please consider that area values and specific waste factors can be influenced by the size and complexity of the property, captured image quality, specific siding techniques, and your own level of expertise. Accessories are not included in these values and may require additional material.

These tables are only intended to make common waste calculations easier and should not be interpreted as recommendations.

Windows

Group	Group Width x Height	Group United Inches	Opening	Width x Height	United Inches	Area
WG-1	33" x 62"	94"	W-101	33" x 62"	94"	14 ft ²
WG-2	33" x 62"	94"	W-102	33" x 62"	94"	14 ft ²
WG-3	33" x 62"	94"	W-103	33" x 62"	94"	14 ft ²
WG-4	33" x 62"	94"	W-104	33" x 62"	94"	14 ft ²
WG-5	33" x 62"	94"	W-105	33" x 62"	94"	14 ft ²
WG-6	33" x 62"	94"	W-106	33" x 62"	94"	14 ft ²
WG-7	33" x 62"	94"	W-107	33" x 62"	94"	14 ft ²
WG-8	33" x 62"	94"	W-108	33" x 62"	94"	14 ft ²
WG-9	33" x 62"	94"	W-109	33" x 62"	94"	14 ft ²
WG-10	33" x 62"	94"	W-110	33" x 62"	94"	14 ft ²
WG-11	33" x 62"	94"	W-111	33" x 62"	94"	14 ft ²
WG-12	33" x 62"	94"	W-112	33" x 62"	94"	14 ft ²
WG-13	33" x 62"	94"	W-113	33" x 62"	94"	14 ft ²
WG-14	33" x 62"	94"	W-114	33" x 62"	94"	14 ft ²
WG-15	33" x 62"	94"	W-115	33" x 62"	94"	14 ft ²
WG-16	33" x 62"	94"	W-116	33" x 62"	94"	14 ft ²
WG-17	33" x 62"	94"	W-117	33" x 62"	94"	14 ft ²
WG-18	33" x 62"	94"	W-118	33" x 62"	94"	14 ft ²

Windows (cont.)

Group	Group Width x Height	Group United Inches	Opening	Width x Height	United Inches	Area
WG-19	33" x 62"	94"	W-119	33" x 62"	94"	14 ft ²
WG-20	33" x 62"	94"	W-120	33" x 62"	94"	14 ft ²
WG-21	33" x 62"	94"	W-121	33" x 62"	94"	14 ft ²
WG-22	33" x 62"	94"	W-122	33" x 62"	94"	14 ft ²
WG-23	33" x 62"	94"	W-123	33" x 62"	94"	14 ft ²
WG-24	33" x 62"	94"	W-124	33" x 62"	94"	14 ft ²
WG-25	33" x 62"	94"	W-125	33" x 62"	94"	14 ft ²
WG-26	33" x 62"	94"	W-126	33" x 62"	94"	14 ft ²
WG-27	33" x 62"	94"	W-127	33" x 62"	94"	14 ft ²
WG-28	33" x 62"	94"	W-128	33" x 62"	94"	14 ft ²
WG-29	33" x 62"	94"	W-129	33" x 62"	94"	14 ft ²
			Total	-	2738"	406 ft ²

Doors

Opening	Width x Height
D-1	36" x 80"
D-2	36" x 80"
D-3	36" x 80"
D-4	36" x 80"
D-5	36" x 80"

*Door height and width have been snapped to standard

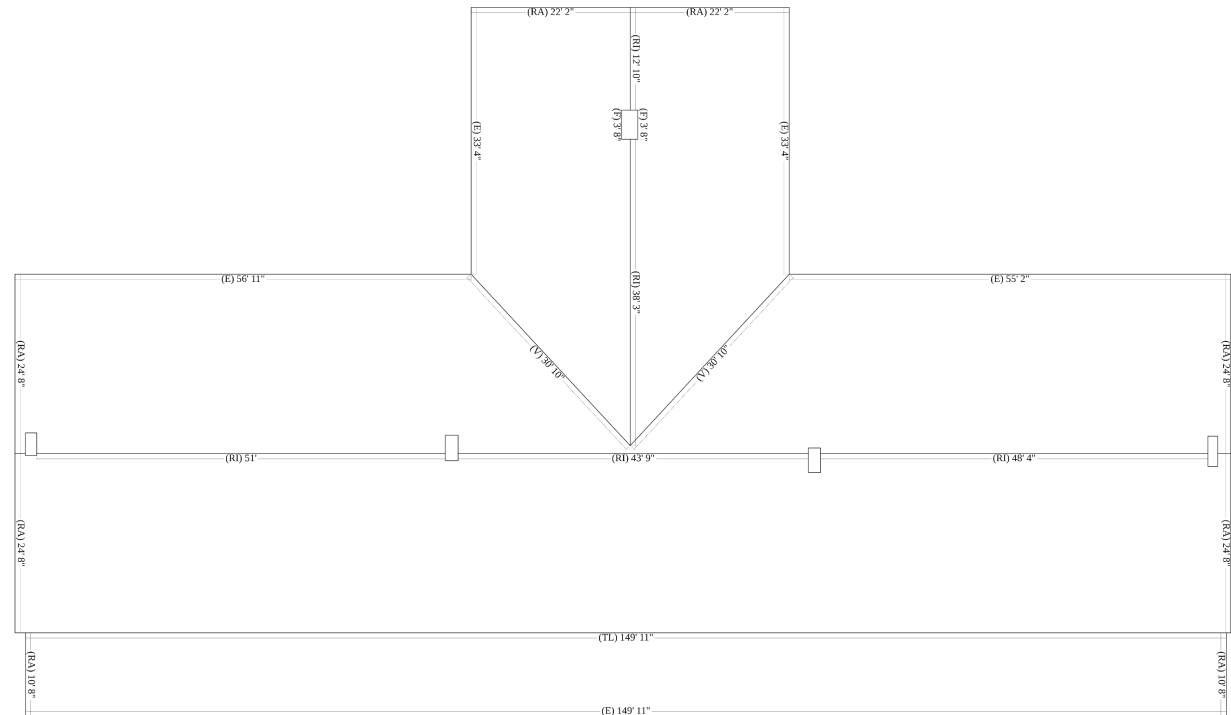
Entire Doors

Opening	Width x Height	Area
D-1	44" x 94"	29 ft ²
D-2	44" x 94"	29 ft ²
D-3	36" x 76"	19 ft ²
D-4	35" x 92"	22 ft ²
D-5	36" x 77"	20 ft ²
Total	-	119 ft ²

*Total door square footage includes entire door package (e.g. with transoms, sidelites, etc.)

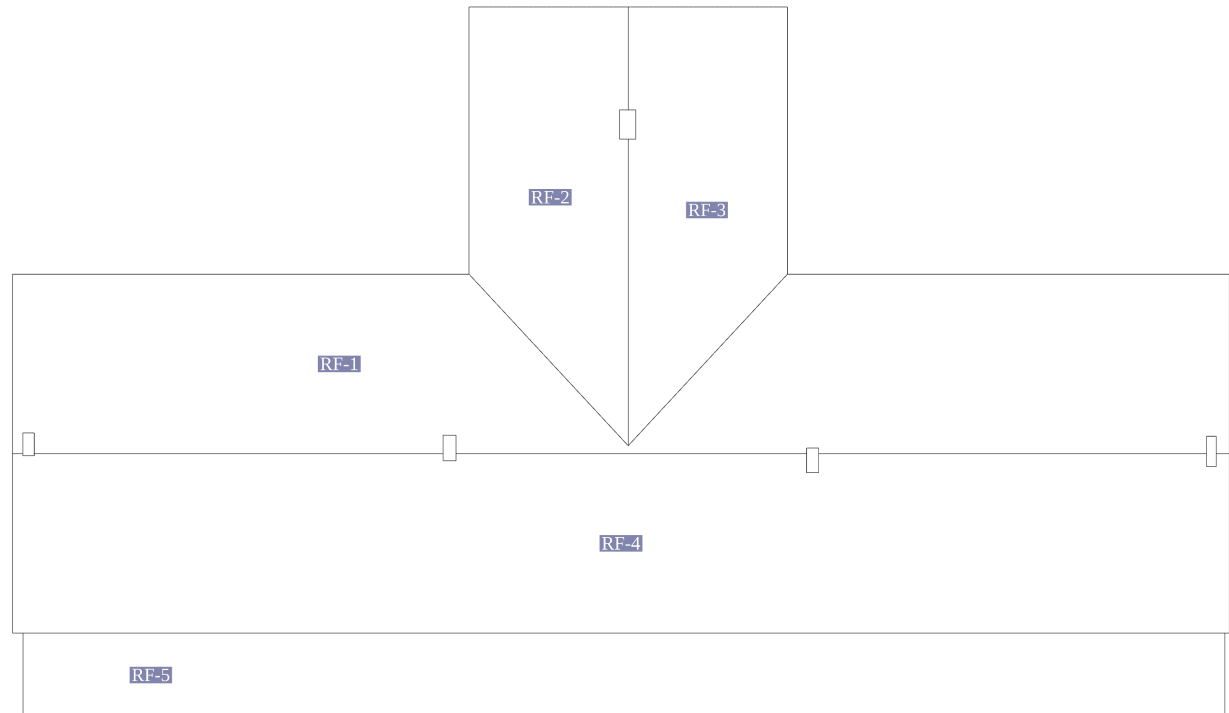
Roof	Length
Ridges (RI)	197' 1"
Hips (H)	-
Valleys (V)	61' 8"
Rakes (RA)	164' 4"
Eaves (E)	330' 5"
Flashing (F)*	18' 9"
Step Flashing (SF)*	32' 11"
Transition Line (TL)	149' 11"

*Please view the 3D model for more detail (e.g. flashing, step flashing and some other roof lines may be difficult to see on the PDF)

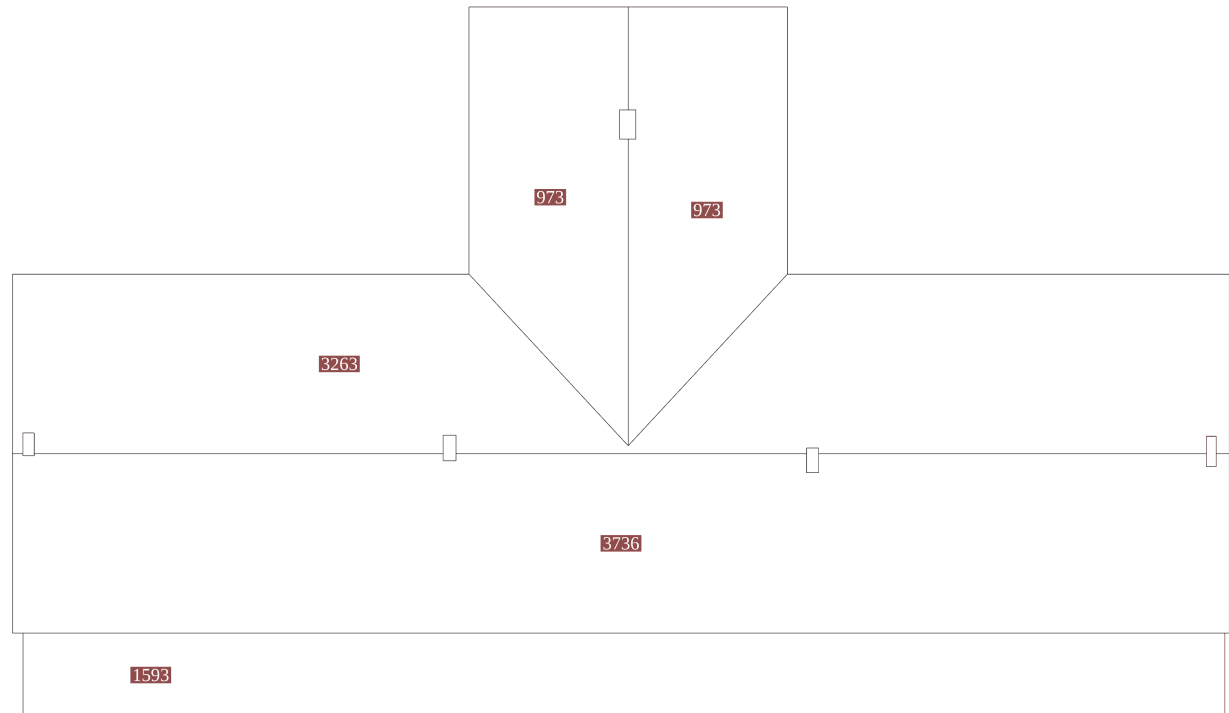


Roof Facets

Facet	Area	Pitch
RF-1	3263 ft ²	6/12
RF-2	973 ft ²	6/12
RF-3	973 ft ²	6/12
RF-4	3736 ft ²	6/12
RF-5	1593 ft ²	2/12



Roof	Facets	Total
Total	5	10538 ft ²



Roof Pitch	Area	Percentage
6 / 12	8945 ft ²	84.88%
2 / 12	1593 ft ²	15.12%

